



COMPREHENSIVE WIRELESS INFRASTRUCTURE SECURITY ASSESSMENT, ATTACK SIMULATION, AND REMEDIATION ANALYSIS

Prepared For

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1. Executive Summary

This report presents the results of a comprehensive wireless infrastructure security assessment, attack simulation, and remediation analysis conducted as part of the Ethical Hacking Professional Course.

The assessment was performed within a controlled and authorized testing environment to evaluate wireless security risks, identify attack paths, and assess the effectiveness of existing wireless security controls. The project was based on the enterprise scenario of KAPAS INTERNATIONAL and utilized a real-world wireless environment to simulate enterprise wireless reconnaissance, monitoring, attack simulation, and security investigation activities.

The primary assessment target selected during testing was the wireless network identified as BET9A@IKAY SHOP. This wireless environment served as the basis for wireless reconnaissance, security testing, packet analysis, attack simulation, and risk assessment activities documented throughout this report.

The assessment focused on identifying weaknesses associated with wireless authentication, wireless encryption, unauthorized wireless access, credential exposure, rogue access points, insecure wireless configurations, weak network segmentation, and wireless monitoring limitations.

Industry-standard wireless assessment tools including Airodump-ng, Airededdon, and Wireshark were utilized to perform wireless discovery, packet capture, handshake collection, traffic analysis, and attack simulation activities.

The assessment successfully demonstrated wireless reconnaissance techniques, WPA2 handshake and PMKID capture capabilities, wireless traffic monitoring, packet analysis, Evil Twin attack deployment, and offline password attack testing.

The findings obtained during the assessment were mapped against MITRE ATT&CK techniques to provide visibility into potential attacker behavior and enterprise wireless security risks.

Recommendations have been provided to improve wireless authentication mechanisms, strengthen monitoring and detection capabilities, implement network segmentation controls, and support the development of a secure enterprise wireless architecture.

All assessment activities documented within this report were conducted for educational, defensive security assessment, and authorized cybersecurity training purposes only.

2. Project Objectives

The primary objective of this project was to assess the wireless security posture of a simulated enterprise environment based on KAPAS INTERNATIONAL.

Specific objectives included:

- Identification of available wireless networks and hidden SSIDs
- Analysis of WPA, WPA2, and WPA3 wireless security mechanisms
- Identification of weak wireless configurations
- Detection of rogue wireless devices and unauthorized access points
- Wireless reconnaissance using industry-standard assessment tools
- Simulation of real-world wireless attack techniques
- Evaluation of guest wireless and BYOD security risks
- Assessment of wireless segmentation effectiveness
- Mapping of identified risks to MITRE ATT&CK techniques
- Development of enterprise wireless security recommendations
- Design of a secure wireless architecture
- Development of wireless monitoring and incident response strategies

3. Assessment Scope

The assessment focused on wireless infrastructure components and wireless-connected environments associated with enterprise operations.

The scope included:

- Wireless network discovery
- SSID identification
- Hidden SSID detection
- BSSID identification
- Channel analysis
- Signal strength analysis
- Wireless client observation
- WPA2 security assessment
- Handshake capture testing
- PMKID assessment
- Evil Twin attack simulation
- Offline password attack testing
- Wireless packet analysis
- MITRE ATT&CK mapping
- Wireless risk assessment

Additional wireless devices observed during reconnaissance activities included guest wireless networks, mobile hotspots, consumer wireless devices, wireless printers, and enterprise-style wireless deployments.

4. Rules of Engagement

This assessment was conducted under controlled and authorized conditions as part of a wireless security evaluation exercise.

The following rules of engagement were observed throughout the assessment:

- Testing activities were limited to authorized environments
- Wireless assessment activities were conducted strictly for educational purposes
- No destructive actions were performed
- No unauthorized access to sensitive information was attempted
- Service disruption activities were minimized
- Evidence collection procedures were followed throughout the assessment
- Findings were documented responsibly and professionally
- All attack simulations were conducted within authorized testing boundaries

5. Assessment Methodology

The assessment followed a structured wireless security testing methodology designed to simulate real-world attacker behavior while maintaining safe and controlled testing conditions.

The methodology consisted of the following phases:

1. Planning and Scope Definition
2. Wireless Reconnaissance
3. Wireless Security Analysis
4. Wireless Attack Simulation
5. Evidence Collection and Validation
6. Wireless Traffic Monitoring and Analysis
7. MITRE ATT&CK Mapping
8. Risk Assessment
9. Remediation Planning
10. Security Architecture Review

6. Wireless Reconnaissance Planning

A controlled wireless assessment environment was prepared to support wireless reconnaissance, packet capture, monitoring, and attack simulation activities.

Hardware Components

- Laptop systems running Kali Linux
- External wireless adapters supporting monitor mode
- Authorized wireless assessment target

Software and Tools

Airodump-ng

Used to identify wireless networks, discover hidden SSIDs, enumerate BSSIDs, analyze wireless channels, observe signal strength, and monitor client activity.

Airgeddon

Used to manage monitor mode, perform wireless assessment activities, capture handshakes, collect PMKIDs, and simulate wireless attack scenarios.

Wireshark

Used for packet capture, protocol analysis, wireless traffic investigation, and packet-level inspection.

Kali Linux Wireless Utilities

Used for interface management and monitor mode configuration.

7. Evidence Collection & Documentation Plan

Evidence collection procedures were established prior to beginning practical assessment activities.

Throughout the assessment, screenshots, packet captures, command outputs, traffic analysis results, handshake captures, PMKID captures, and attack simulation outputs were collected and documented to support assessment findings.

Evidence collected included:

- Wireless adapter configuration
- Monitor mode activation
- Wireless reconnaissance outputs
- SSID discovery results
- Hidden network identification
- Signal strength analysis
- Channel allocation analysis
- Handshake capture evidence
- PMKID capture evidence
- Evil Twin attack simulation outputs
- Offline password attack results
- Wireshark packet analysis
- Traffic monitoring observations
- Security findings and recommendations

All evidence was documented and retained to support findings presented throughout this report.

8. Assessment Target Selection

A wireless reconnaissance scan was performed to identify available wireless networks within the assessment environment.

Following the reconnaissance phase, the wireless network selected for detailed assessment was:

SSID: BET9A@IKAY SHOP

BSSID: B4:0F:3B:2C:8B

Channel: 8

Frequency Band: 2.4 GHz

Encryption Type: WPA2-Personal (WPA2-PSK)

Signal Strength: Approximately 35%

Connected Clients Observed: Yes

The selected wireless network provided sufficient visibility for wireless reconnaissance, packet capture, handshake assessment, PMKID analysis, attack simulation, and wireless traffic monitoring activities.

9. Monitor Mode Configuration

Monitor mode was configured on the wireless adapter to enable wireless packet capture, wireless reconnaissance, and security assessment activities.

Unlike managed mode, monitor mode allows a wireless adapter to capture all wireless frames within range without associating with a specific access point. This capability is essential for wireless penetration testing, packet analysis, handshake collection, client discovery, and wireless attack simulation.

Prior to commencing reconnaissance activities, the wireless adapter was verified to support monitor mode operation and was successfully configured using Aircgeddon and Kali Linux wireless utilities.

WIRELESS ADAPTER IDENTIFICATION

The assessment began with identification and verification of the wireless adapter connected to the assessment workstation.

The wireless adapter detected during the assessment was:

Interface: wlan0

Chipset: Qualcomm Atheros AR9271

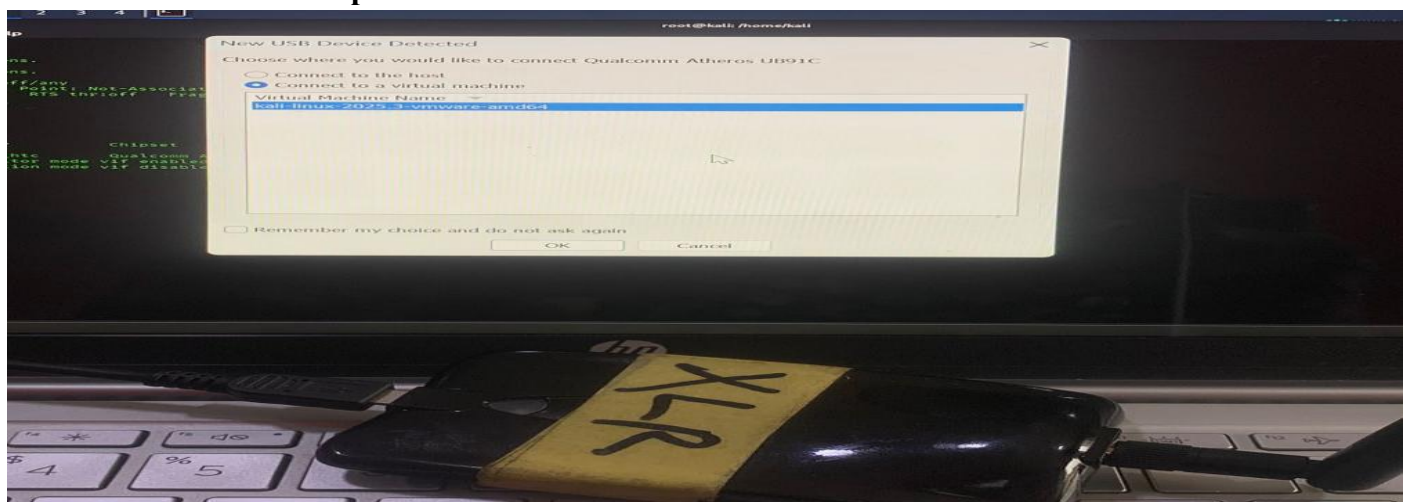
Wireless Standard: IEEE 802.11n

Frequency Band: 2.4 GHz

Monitor Mode Support: Supported

The adapter was successfully recognized by the operating system and was available for wireless assessment activities.

Screenshot - wireless-adapter-detection.



Filename: 01-wireless-adapter-detection.png

```
Session Actions Edit View Help
root@kali: ~
# iwconfig
lo      no wireless extensions.

eth0    no wireless extensions.

wlan0    IEEE 802.11  ESSID:off/any
          Mode:Managed  Access Point: Not-Associated  Tx-Power=0 dBm
          Retry short limit:7   RTS thr:off   Fragment thr:off
          Encryption key:off
          Power Management:off

root@kali: ~
# airmon-ng

PHY      Interface      Driver      Chipset
phy0     wlan0              ath9k_htc   Qualcomm Atheros Communications AR9271 802.11n
```

Filename: 01.1-wireless-adapter-detection.png

Description:

Screenshot showing successful detection of the external wireless adapter and verification of wireless hardware prior to monitor mode activation.

WIRELESS INTERFACE SELECTION

Following adapter verification, Airedgdon was launched and used to identify available network interfaces.

The wireless assessment interface was selected from the available interfaces presented by the tool.

The selected interface was:

Interface: wlan0mon

Chipset: Qualcomm Atheros AR9271

Supported Band: 2.4 GHz

Mode: Wireless Assessment Interface

Screenshot – Interface Selection

```
Session Actions Edit View Help
root@kali: ~
***** Interface selection *****
Select an interface to work with:

1. eth0 // Chipset: Intel Corporation 82545EM
2. wlan0mon // 2.4Ghz (WiFi4) // Chipset: Qualcomm Atheros Communications AR9271 802.11n

Hint If you have any doubt or problem, you can check Wiki FAQ section (https://github.com/v1s1t0r1sh3r3/airgeddon/wiki/FAQ%20%20Troubleshooting) or ask in our Discord server: https://discord.gg/sQ9dgt9

> 2
```

Filename: 02-interface-selection.png

Description:

Screenshot showing Airgeddon interface selection menu and selection of the wireless adapter for assessment activities.

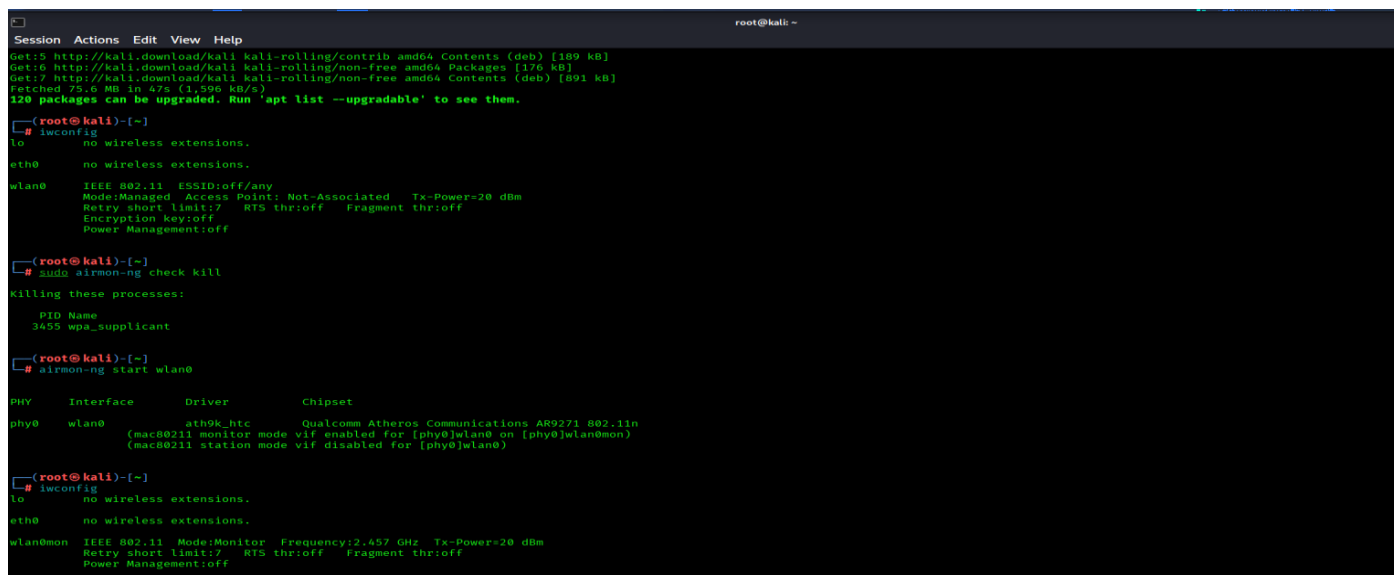
MONITOR MODE ACTIVATION

After interface selection, the wireless adapter was transitioned from managed mode to monitor mode.

Monitor mode activation was required to allow passive observation of wireless traffic, beacon frames, management frames, authentication exchanges, and other wireless communications occurring within range of the assessment system.

Airgeddon successfully configured the adapter and verified monitor mode operation.

Screenshot – Monitor Mode Enabled



```
root@kali: ~  
Session Actions Edit View Help  
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [189 kB]  
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [176 kB]  
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [891 kB]  
Fetched 25.6 MB in 47s (1,350 kB/s)  
120 packages can be upgraded. Run 'apt list --upgradable' to see them.  
  
(root@kali)-[~]  
# iwconfig  
lo        no wireless extensions.  
  
eth0      no wireless extensions.  
  
wlan0     IEEE 802.11 ESSID:off/any  
          Mode:Managed Access Point: Not-Associated Tx-Power=20 dBm  
          Retry short limit:7 RTS thr:off Fragment thr:off  
          Encryption key:off  
          Power Management:off  
  
(root@kali)-[~]  
# sudo airmon-ng check kill  
Killing these processes:  
  
PID Name  
3455 wpa_supplicant  
  
(root@kali)-[~]  
# airmon-ng start wlan0  
  
PHY      Interface  Driver      Chipset  
phy0     wlan0      ath9k_htc   Qualcomm Atheros Communications AR9271 802.11n  
          (mac80211 monitor mode vif enabled for [phy0]wlan0 on [phy0]wlan0mon)  
          (mac80211 station mode vif disabled for [phy0]wlan0)  
  
(root@kali)-[~]  
# iwconfig  
lo        no wireless extensions.  
  
eth0      no wireless extensions.  
  
wlan0mon  IEEE 802.11 Mode:Monitor Frequency:2.457 GHz Tx-Power=20 dBm  
          Retry short limit:7 RTS thr:off Fragment thr:off  
          Power Management:off
```

Filename: 03-monitor-mode-enabled.png

Description:

Screenshot showing successful activation of monitor mode and confirmation that the wireless adapter was ready for wireless reconnaissance activities.

Assessment Result

Monitor mode was successfully enabled and verified.

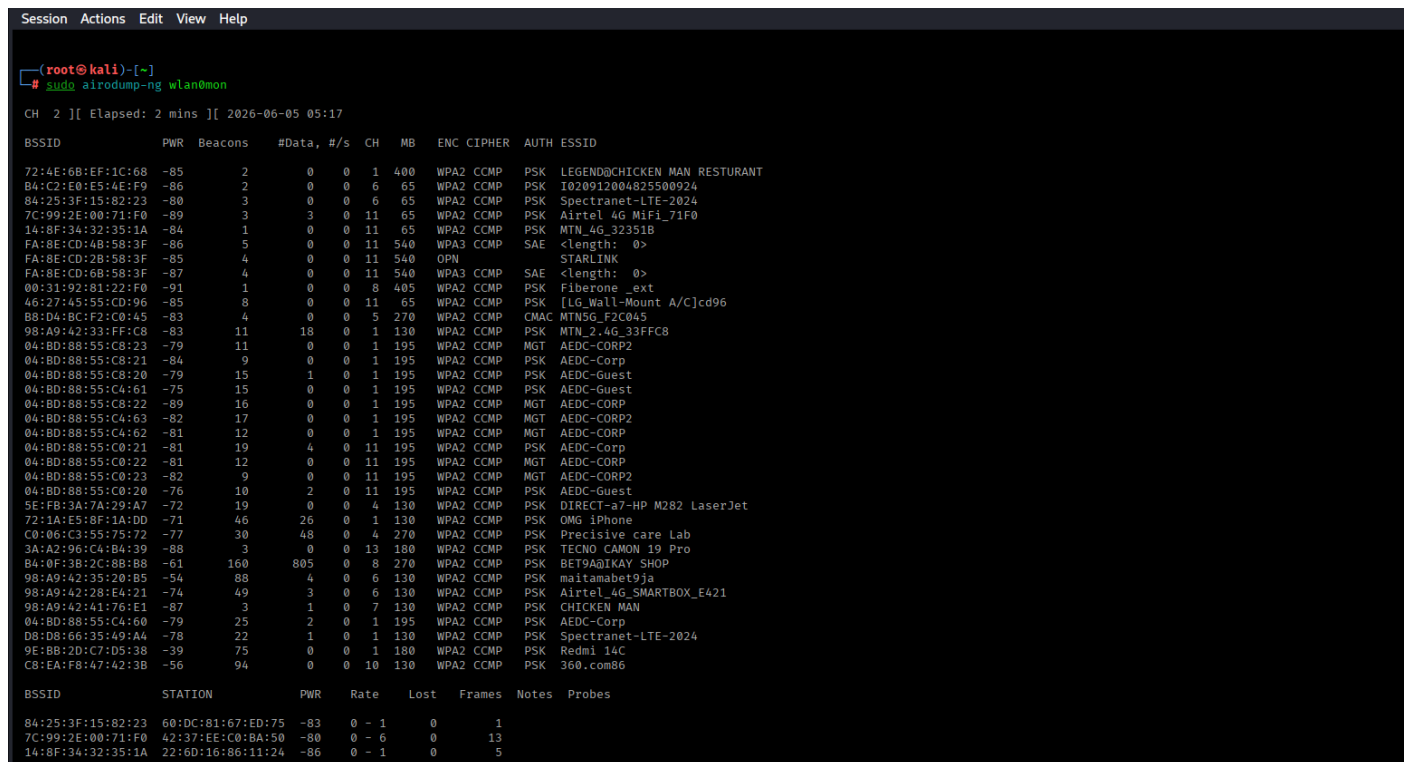
The wireless adapter was fully operational and capable of performing wireless reconnaissance, packet capture, handshake collection, and wireless traffic analysis.

10. RF Reconnaissance Results

RF (Radio Frequency) reconnaissance was conducted to identify wireless networks operating within the assessment area. The objective was to collect information relating to wireless network names, access point identifiers, encryption mechanisms, channel allocations, signal strengths, and potential rogue devices.

The reconnaissance phase was performed using Airodump-ng while the wireless adapter was operating in monitor mode.

Screenshot – Wireless Reconnaissance Scan Results



The screenshot shows the Airodump-ng terminal output. At the top, it displays the session title 'Session Actions Edit View Help' and the command 'sudo airodump-ng wlan0mon'. Below this, it shows the channel 'CH 2' and the elapsed time '2 mins' as of '2026-06-05 05:17'. The main table lists various wireless networks with columns for BSSID, PWR, Beacons, #Data, #/s, CH, MB, ENC, CIPHER, AUTH, and ESSID. The networks listed include 'LEGEND@CHICKEN MAN RESTURANT', 'I020912004825500924', 'Spectranet-LTE-2024', 'Airtel 4G MiFi_71f0', 'MTN_4G_32351B', 'SAE <length: 0>', 'STARLINK', 'Fiberone_ext', '[LG_Wall-Mount A/C]cd96', 'CMAC MTN5G_F2C045', 'MTN_2.4G_33FFC8', 'AEDC-CORP2', 'AEDC-Corp', 'AEDC-Guest', 'AEDC-CORP', 'AEDC-CORP2', 'AEDC-CORP', 'AEDC-CORP2', 'AEDC-Guest', 'DIRECT-a7-HP M282 LaserJet', 'OMG iPhone', 'Precisive care Lab', 'TECNO CAMON 19 Pro', 'BET9A@IKAY SHOP', 'maitamabet9ja', 'Airtel_4G_SMARTBOX_E421', 'CHICKEN MAN', 'AEDC-Corp', 'Spectranet-LTE-2024', 'Redmi 14C', and '360.com86'. Below the main table, there is a summary table with columns for BSSID, STATION, PWR, Rate, Lost, Frames, Notes, and Probes, showing data for three specific networks.

BSSID	PWR	Beacons	#Data	#/s	CH	MB	ENC	CIPHER	AUTH	ESSID
72:4E:6B:EF:1C:68	-85	2	0	0	1	400	WPA2	CCMP	PSK	LEGEND@CHICKEN MAN RESTURANT
B4:C2:E0:E5:4E:F9	-86	2	0	0	6	65	WPA2	CCMP	PSK	I020912004825500924
84:25:3F:15:82:23	-80	3	0	0	6	65	WPA2	CCMP	PSK	Spectranet-LTE-2024
7C:99:2E:00:71:F0	-89	3	3	0	11	65	WPA2	CCMP	PSK	Airtel 4G MiFi_71f0
14:8F:34:32:35:1A	-84	1	0	0	11	65	WPA2	CCMP	PSK	MTN_4G_32351B
FA:8E:CD:4B:58:3F	-86	5	0	0	11	540	WPA3	CCMP	SAE <length: 0>	
FA:8E:CD:2B:58:3F	-85	4	0	0	11	540	OPN			STARLINK
FA:8E:CD:6B:58:3F	-87	4	0	0	11	540	WPA3	CCMP	SAE <length: 0>	
00:31:92:81:22:F0	-91	1	0	0	8	405	WPA2	CCMP	PSK	Fiberone_ext
46:27:45:55:CD:96	-85	8	0	0	11	65	WPA2	CCMP	PSK	[LG_Wall-Mount A/C]cd96
BB:D4:BC:F2:C0:45	-83	4	0	0	5	270	WPA2	CCMP	CMAC	MTN5G_F2C045
9B:A9:42:33:FF:C8	-83	11	18	0	1	130	WPA2	CCMP	PSK	MTN_2.4G_33FFC8
04:BD:88:55:C8:23	-79	11	0	0	1	195	WPA2	CCMP	MGT	AEDC-CORP2
04:BD:88:55:C8:21	-84	9	0	0	1	195	WPA2	CCMP	PSK	AEDC-Corp
04:BD:88:55:C8:20	-79	15	1	0	1	195	WPA2	CCMP	PSK	AEDC-Guest
04:BD:88:55:C4:61	-75	15	0	0	1	195	WPA2	CCMP	PSK	AEDC-Guest
04:BD:88:55:C8:22	-89	16	0	0	1	195	WPA2	CCMP	MGT	AEDC-CORP
04:BD:88:55:C4:63	-82	17	0	0	1	195	WPA2	CCMP	MGT	AEDC-CORP2
04:BD:88:55:C4:62	-81	12	0	0	1	195	WPA2	CCMP	MGT	AEDC-CORP
04:BD:88:55:C0:21	-81	19	4	0	11	195	WPA2	CCMP	PSK	AEDC-Corp
04:BD:88:55:C0:22	-81	12	0	0	11	195	WPA2	CCMP	MGT	AEDC-CORP
04:BD:88:55:C0:23	-82	9	0	0	11	195	WPA2	CCMP	MGT	AEDC-CORP2
04:BD:88:55:C0:20	-76	10	2	0	11	195	WPA2	CCMP	PSK	AEDC-Guest
5E:FB:3A:7A:29:A7	-72	19	0	0	4	130	WPA2	CCMP	PSK	DIRECT-a7-HP M282 LaserJet
72:1A:E5:0F:1A:00	-71	46	26	0	1	130	WPA2	CCMP	PSK	OMG iPhone
C0:06:C3:55:75:72	-77	30	48	0	4	270	WPA2	CCMP	PSK	Precisive care Lab
3A:A2:96:C4:B4:39	-88	3	0	0	13	180	WPA2	CCMP	PSK	TECNO CAMON 19 Pro
B4:0F:38:2C:8B:B8	-61	160	805	0	8	270	WPA2	CCMP	PSK	BET9A@IKAY SHOP
9B:A9:42:35:20:B5	-54	88	4	0	6	130	WPA2	CCMP	PSK	maitamabet9ja
9B:A9:42:28:E4:21	-74	49	3	0	6	130	WPA2	CCMP	PSK	Airtel_4G_SMARTBOX_E421
9B:A9:42:41:76:E1	-87	3	1	0	7	130	WPA2	CCMP	PSK	CHICKEN MAN
04:BD:88:55:C4:60	-79	25	2	0	1	195	WPA2	CCMP	PSK	AEDC-Corp
D8:D8:66:35:49:A4	-78	22	1	0	1	130	WPA2	CCMP	PSK	Spectranet-LTE-2024
9E:BB:20:C7:D5:38	-39	75	0	0	1	180	WPA2	CCMP	PSK	Redmi 14C
C8:EA:F8:47:42:3B	-56	94	0	0	10	130	WPA2	CCMP	PSK	360.com86

BSSID	STATION	PWR	Rate	Lost	Frames	Notes	Probes
84:25:3F:15:82:23	60:DC:81:67:ED:75	-83	0 - 1	0	1		
7C:99:2E:00:71:F0	42:37:EE:C0:BA:50	-80	0 - 6	0	13		
14:8F:34:32:35:1A	22:6D:16:B6:11:24	-86	0 - 1	0	5		

Filename: 04-target-selection.png

Airodump-ng was used to perform passive wireless reconnaissance across the assessment area. Multiple wireless networks were identified, including visible and hidden SSIDs. Information collected included BSSIDs, encryption types, channels, signal strengths, and client activity.

Reconnaissance Findings

- Multiple wireless networks identified
- Hidden SSIDs detected
- WPA and WPA2 networks observed
- Active wireless clients detected
- Several candidate target networks identified

Target Network Identification

After reviewing available wireless networks, the target network was selected for further assessment.

The selected wireless network was:

Parameter	Value
SSID	BET9A@IKAY SHOP
BSSID	B4:0F:3B:2C:8B:B8
Channel	8
Encryption	WPA2
Authentication	PSK
Cipher	CCMP (AES)
Signal Strength	-32 dBm

A focused scan was then performed against the target BSSID using Airodump-ng.

The scan revealed multiple connected wireless clients communicating with the access point. During monitoring, EAPOL traffic was observed, indicating authentication activity between wireless clients and the access point.

Additionally, a WPA handshake indicator was displayed by Airodump-ng, confirming that authentication traffic had been captured during the assessment.

Screenshot – Target Network Analysis

Session Actions Edit View Help										
CH 8][Elapsed: 2 mins][2026-06-05 05:26][WPA handshake: B4:0F:3B:2C:8B:B8										
BSSID	PWR	RXQ	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
B4:0F:3B:2C:8B:B8	-32	0	1530	27101 224	8	270	WPA2	CCMP	PSK	BET9A@IKAY SHOP
BSSID	STATION		PWR	Rate	Lost	Frames	Notes	Probes		
B4:0F:3B:2C:8B:B8	00:1F:3B:BD:6D:C3		-42	5e- 6e	0	266				
B4:0F:3B:2C:8B:B8	00:1B:77:7F:8F:55		-57	54e-48e	1	2401				
B4:0F:3B:2C:8B:B8	00:1B:9E:C9:34:57		-51	54 -54	0	341				
B4:0F:3B:2C:8B:B8	58:C5:87:A7:29:21		-45	1e- 1	85	258	EAPOL			
B4:0F:3B:2C:8B:B8	00:16:44:9F:81:4D		-51	54 -54	3	3637				
B4:0F:3B:2C:8B:B8	00:1B:77:26:1E:60		-54	54e-54e	1	1201				
B4:0F:3B:2C:8B:B8	48:22:54:1C:A4:B9		-70	6e- 6e	460	436				
B4:0F:3B:2C:8B:B8	00:13:E8:2F:8D:7F		-59	1e- 5e	0	98				
B4:0F:3B:2C:8B:B8	60:6C:66:55:51:0E		-52	24e- 6e	0	18660				

Filename: 04.1-target-selection.png

Reconnaissance Findings

The following observations were made during the RF reconnaissance phase:

- Target wireless network was actively broadcasting its SSID.
- WPA2-PSK authentication was in use.
- CCMP (AES) encryption was enabled.
- Multiple client devices were connected.
- Authentication traffic (EAPOL) was observed.
- WPA handshake capture was detected.
- No hidden SSID was identified for the selected target.
- The wireless network was operating on Channel 8 within the 2.4 GHz frequency band.

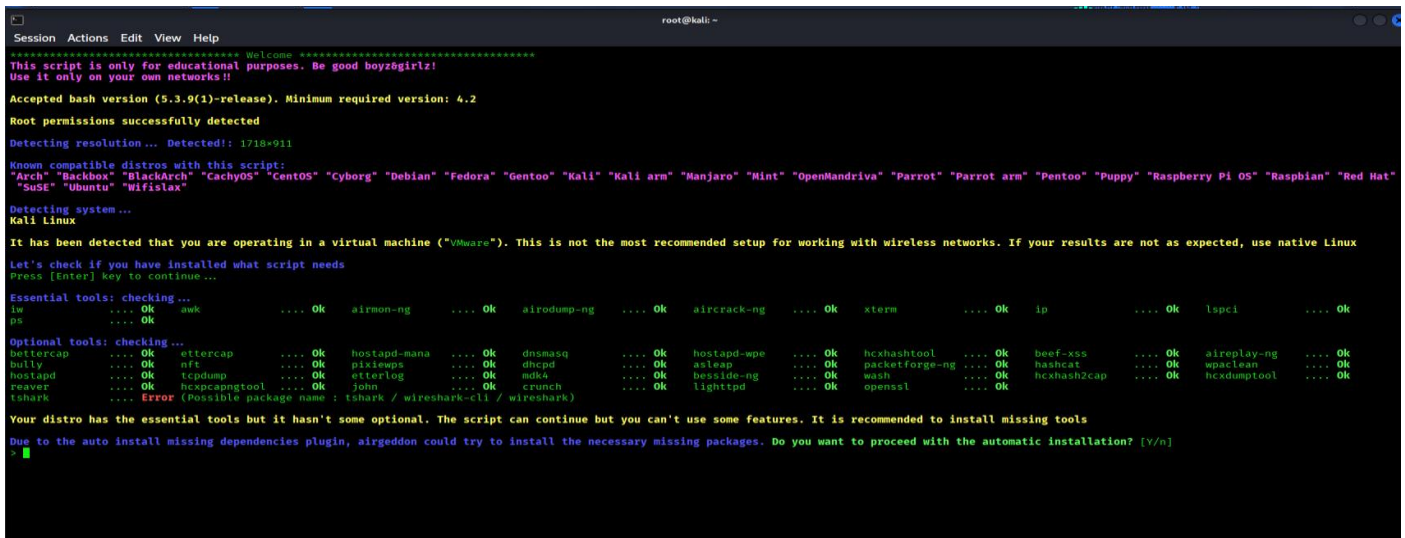
The information gathered during this phase provided the foundation for subsequent wireless attack simulation, handshake analysis, and packet inspection activities.

11. Wireless Attack Chain Analysis

Airgeddon was used during the assessment phase to perform wireless attack simulations against the target wireless network **BET9A@IKAY SHOP**. The tool was utilized to validate wireless security controls, assess WPA2 authentication resilience, evaluate handshake capture exposure, and observe potential risks associated with rogue access point attacks and offline password attacks.

The following assessment stages were performed.

Screenshot – Airgeddon Main Menu

A terminal window titled 'root@kali ~' showing the Airgeddon main menu. The menu includes a welcome message, a warning to use it only on one's own networks, and a list of supported Linux distributions. It then checks for essential tools (iw, iwconfig, aircrack-ng, airodump-ng, xterm, ip, lspci) and optional tools (ettercap, hostapd-mana, dnsmasq, hostapd-wpe, hcxhashtool, beef-xss, aireplay-ng, etc.). A message indicates that while essential tools are present, some optional ones are missing, and it asks if the user wants to proceed with automatic installation of missing packages. The user has entered 'y' for yes.

```
root@kali ~
Session Actions Edit View Help
***** Welcome *****
This script is only for educational purposes. Be good boys&girls!
Use it only on your own networks!!

Accepted bash version (5.3.9(1)-release). Minimum required version: 4.2
Root permissions successfully detected
Detecting resolution... Detected!: 1718x911

Known compatible distros with this script:
"Arch" "Backbox" "BlackArch" "CachyOS" "CentOS" "Cyborg" "Debian" "Fedora" "Gentoo" "Kali" "Kali arm" "Manjaro" "Mint" "OpenMandriva" "Parrot" "Parrot arm" "Pentoo" "Puppy" "Raspbian" "Red Hat"
"SUSE" "Ubuntu" "Wifislax"

Detecting system...
Kali Linux

It has been detected that you are operating in a virtual machine ("VMware"). This is not the most recommended setup for working with wireless networks. If your results are not as expected, use native Linux

Let's check if you have installed what script needs
Press [Enter] key to continue...

Essential tools: checking...
iw      .... Ok
iwconfig .... Ok
aircrack-ng .... Ok
airodump-ng .... Ok
xterm   .... Ok
ip       .... Ok
lspci    .... Ok

Optional tools: checking...
ettercap .... Ok
hostapd-mana .... Ok
dnsmasq  .... Ok
hostapd-wpe .... Ok
hcxhashtool .... Ok
beef-xss  .... Ok
aireplay-ng .... Ok
bully     .... Ok
nft       .... Ok
pihole    .... Ok
dnsmasq   .... Ok
asleap    .... Ok
packetforge-ng .... Ok
hashcat   .... Ok
wpacli    .... Ok
hostapd   .... Ok
tcpdump   .... Ok
etterlog  .... Ok
mdk3      .... Ok
besside-ng .... Ok
wash      .... Ok
hcxhash2cap .... Ok
hcxdump   .... Ok
reaver    .... Ok
hcxpcapngtool .... Ok
john      .... Ok
crunch     .... Ok
lighttpd  .... Ok
openssl   .... Ok
tshark     .... Error (Possible package name : tshark / wireshark-cli / wireshark)

Your distro has the essential tools but it hasn't some optional. The script can continue but you can't use some features. It is recommended to install missing tools
Due to the auto install missing dependencies plugin, airgeddon could try to install the necessary missing packages. Do you want to proceed with the automatic installation? [Y/n]
> y
```

Filename: 05-airgeddon-main-menu.png

Description

The screenshot shows the Airgeddon main interface used to access wireless assessment modules.

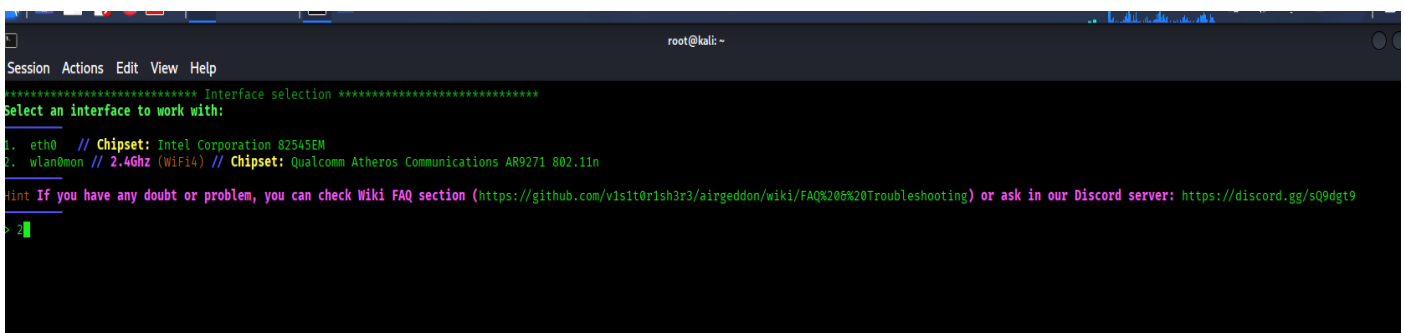
From this interface, wireless attack simulations, handshake capture operations, rogue access point assessments, deauthentication testing, and wireless auditing functions can be initiated.

Observation

The platform provides integrated wireless assessment capabilities for enterprise wireless security evaluations.

Wireless Interface Selection

Screenshot – Interface Selection

A terminal window titled 'root@kali ~' showing the Airgeddon interface selection screen. It prompts the user to 'Select an interface to work with:' and lists two options: 'eth0 // Chipset: Intel Corporation 82545EM' and 'wlan0mon // 2.4Ghz (WiFi4) // Chipset: Qualcomm Atheros Communications AR9271 802.11n'. A hint at the bottom suggests checking the Wiki FAQ section or asking in a Discord server. The user has entered '2' to select wlan0mon.

```
root@kali ~
Session Actions Edit View Help
***** Interface selection *****
Select an interface to work with:

1. eth0 // Chipset: Intel Corporation 82545EM
2. wlan0mon // 2.4Ghz (WiFi4) // Chipset: Qualcomm Atheros Communications AR9271 802.11n

Hint If you have any doubt or problem, you can check Wiki FAQ section (https://github.com/v1s1t0r1sh3r3/airgeddon/wiki/FAQ%20%20Troubleshooting) or ask in our Discord server: https://discord.gg/sQ9dgt9

> 2
```

Filename: 05.1-interface-selection.png

Description

The wireless adapter was selected for assessment activities.

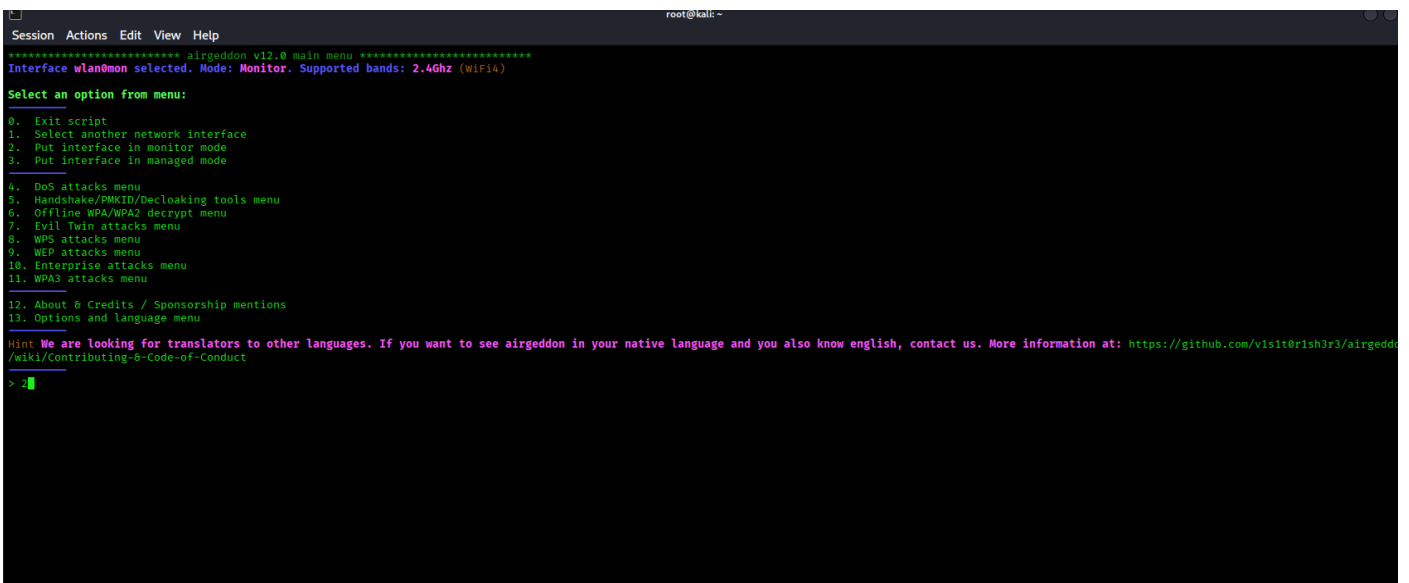
The adapter was configured to support monitor mode operations required for wireless reconnaissance and attack simulation activities.

Observation

The selected wireless interface was successfully recognized and prepared for wireless assessment operations.

Monitor Mode Activation

Screenshot – Monitor Mode Enabled



```
root@kali: ~  
Session Actions Edit View Help  
***** airgeddon v12.0 main menu *****  
Interface wlan0mon selected. Mode: Monitor. Supported bands: 2.4Ghz (WiFi4)  
Select an option from menu:  
0. Exit script  
1. Select another network interface  
2. Put interface in monitor mode  
3. Put interface in managed mode  
4. DoS attacks menu  
5. Handshake/PWKID/Decloaking tools menu  
6. Offline WPA/WPA2 decrypt menu  
7. Evil Twin attacks menu  
8. WPS attacks menu  
9. WEP attacks menu  
10. Enterprise attacks menu  
11. WPA3 attacks menu  
12. About & Credits / Sponsorship mentions  
13. Options and language menu  
Hint We are looking for translators to other languages. If you want to see airgeddon in your native language and you also know english, contact us. More information at: https://github.com/vis1t0rish3r3/airgeddon/wiki/Contributing-to-Code-of-Conduct  
> 2
```

Filename: 05.2-monitor-mode-enabled.png

Description

Monitor mode was enabled on the wireless interface to allow passive capture of wireless management frames, beacon frames, authentication exchanges, and other wireless communications.

Observation

The interface was successfully converted from managed mode to monitor mode and became capable of capturing wireless traffic without joining the network.

Target Network Selection

Screenshot – Airedgdon Target Selection

```
Session Actions Edit View Help
***** Handshake/PMKID/Decloaking tools menu *****
Interface wlan0mon selected. Mode: Monitor. Supported bands: 2.4Ghz (WIFI4)

Select an option from menu:
0. Return to main menu
1. Select another network interface
2. Put interface in monitor mode
3. Put interface in managed mode
4. Explore for targets (monitor mode needed)
5. Capture PMKID
6. Capture Handshake
7. Clean/optimize Handshake file
8. Decloaking by deauthentication
9. (mdk4) Decloaking by dictionary

Hint Another way to decloak a network with a hidden ESSID is to launch a dictionary attack using mdk3/mdk4. All the w...
> 4

There is no valid target network selected. You'll be redirected to select one
Press [Enter] key to continue...

***** Exploring for targets *****
Exploring for targets option chosen (monitor mode needed)

Selected interface wlan0mon is in monitor mode. Exploration can be performed

Chosen action can be carried out only over WPA/WPA2 networks, however WPA3 has been included in the scan filter because these networks sometimes work in "Mixed mode" offering WPA2/WPA3 and in that case they are displayed in the scan window as WPA3. So WPA3 networks will appear but then airedgdon will analyze them after scan to allow you select only those that also offering WPA2

WPA/WPA2/WPA3 filter enabled in scan. When started, press [Ctrl+C] to stop...
Press [Enter] key to continue...

CH 12 [I Elapsed: 30 s [I 2026-06-05 06:32

BSSID PWR Beacons #Data, #/s CH MB ENC CIPHER AUTH ESSID
04:8D:8B:55:C4:80 -77 2 0 0 1 195 WPA2 CCMP PSK HEBC-Corp
C8:1A:F8:4F:42:38 -45 0 0 0 1 150 WPA2 CCMP PSK 360_cowds
04:8D:8B:55:C3:C1 -97 2 0 0 1 195 WPA2 CCMP PSK HEBC-Corp
04:8D:8B:55:C0:21 -80 3 0 0 1 195 WPA2 CCMP PSK HEBC-Corp
B8:42:F7:C8:3B:8D -97 3 0 0 1 130 WPA2 CCMP PSK Dr_Okai_IDU
8E:F8:3A:7A:29:A7 -62 2 0 0 4 150 WPA2 CCMP PSK DIRECTV-77-HP_HSS_LaserJet
C0:9A:17:35:79:72 -81 5 15 0 4 270 WPA2 CCMP PSK Precursive_ses_Lab
72:1A:E5:0F:1A:8D -63 8 0 0 1 130 WPA2 CCMP PSK OHS_iPhone
3e:92:85:C4:84:53 -94 4 0 0 13 180 WPA2 CCMP PSK TENDI_Corom_13_Pro
98:98:42:20:84:21 -61 7 0 0 6 150 WPA2 CCMP PSK Rintel_4G_SmartBox_E421
98:98:42:20:85:85 -47 13 0 0 8 150 WPA2 CCMP PSK natomads3ja
98:98:42:14:17:6E:11 -81 2 0 0 7 130 WPA2 CCMP PSK CHICKEN PSN
9E:18A:10:07:18:58 -81 13 0 0 1 180 WPA2 CCMP PSK Redmi_14C
8E:12:40:58:45:85 -56 13 0 0 1 130 WPA2 CCMP PSK iPhone
98:98:42:23:3F:13 -69 4 22 0 1 130 WPA2 CCMP PSK MTN_2_4G_53FFCB
04:8D:8B:55:C0:22 -82 2 0 0 1 155 WPA2 CCMP ACT HEBC-Corp
B8:1B8:46:135:49:84 -78 1 0 0 1 130 WPA2 CCMP PSK Spectrawat-LTE-2024
04:8D:8B:55:C4:82 -73 4 0 0 1 155 WPA2 CCMP NOT HEBC-Corp
B4:0F:3B:2C:8B:B8 -28 33 180 0 8 270 WPA2 CCMP PSK BET9A@IKAY SHOP

BSSID STATION PWR Rate Lost Frames Notes Probes
(not associated) C6:45:1C:81:0F:109 -88 0 - 1 3 4
```

Filename: 05.3-airgeddon-target-selection.png

Description

The target wireless network was selected from the list of discovered access points.

The selected target was:

Parameter Value

SSID BET9A@IKAY SHOP

BSSID B4:0F:3B:2C:8B:B8

Channel 8

Encryption WPA2-PSK

Cipher CCMP

Observation

The network was successfully loaded into the assessment module and became the focus of subsequent wireless attack simulations.

WPA2 Handshake Capture Assessment

Screenshot – Handshake Capture Attempt

```
Session Actions Edit View Help
root@kali: ~

***** Attack for Handshake *****
Interface wlan0mon selected. Mode: Monitor. Supported bands: 2.4Ghz (WiFi4)
Selected BSSID: B4:0F:3B:2C:8B:B8 (personal)
Selected channel: 8
Selected band: 2.4Ghz
Selected ESSID: BET9A@IKAY SHOP
Type of encryption: WPA2

Select an option from menu:
0. Return to Handshake/PMKID/Deauth tools menu
1. Deauth / disassoc amok mdk4 attack
2. Deauth aireplay attack
3. Auth DoS attack

Hint If you want to learn how to perform professional wireless network assessments, the main author of aircrack-ng recommends the CWP (Certified Wireless Professional) course from aircrack-ng.org/courses/certified-wifichallenge-professional-cwp?ref=c02137

> 1

Type value in seconds (10-100) for timeout or press [Enter] to accept the proposal [20]:
> 100

Timeout set to 100 seconds

Two windows will be opened. One with the Handshake capturer and other with the attack to force clients to reconnect

Don't close any window manually, script will do when needed. In about 100 seconds maximum you'll know if you've got the Handshake
Press [Enter] key to continue ...

Wait. Be patient ...

In addition to capturing a Handshake, it has been verified that a PMKID from the target network has also been successfully captured

Congratulations !!

Type the path to store the file or press [Enter] to accept the default proposal [/root/handshake-B4:0F:3B:2C:8B:B8.cap]
> /home/kali/netcap.cap

The path is valid and you have write permissions. Script can continue ...

Handshake file generated successfully at [/home/kali/netcap.cap]
Press [Enter] key to continue ...
```

Filename: 05.4-handshake-capture-attempt.png

Description

Aircrack-ng initiated a WPA2 handshake capture assessment.

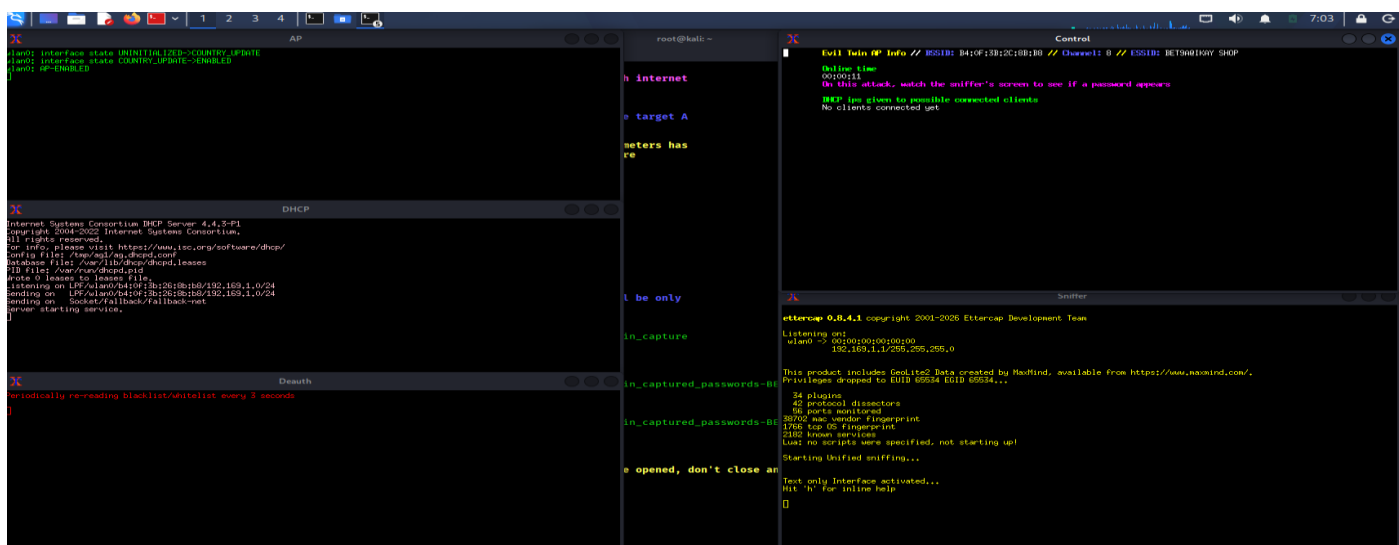
A controlled deauthentication technique was used to encourage wireless clients to reconnect to the access point, thereby generating the WPA2 four-way handshake.

Observation

The assessment process confirmed that connected wireless clients were present and actively communicating with the access point.

Evil Twin Attack Assessment

Screenshot – Evil Twin Attack Menu



Filename: 05.5-evil-twin-attack.png

Description

The Evil Twin attack module was reviewed to assess exposure to rogue access point attacks.

This attack technique involves creating a fraudulent wireless access point that impersonates a legitimate network in order to attract victim devices.

Observation

No clients were connected during testing.

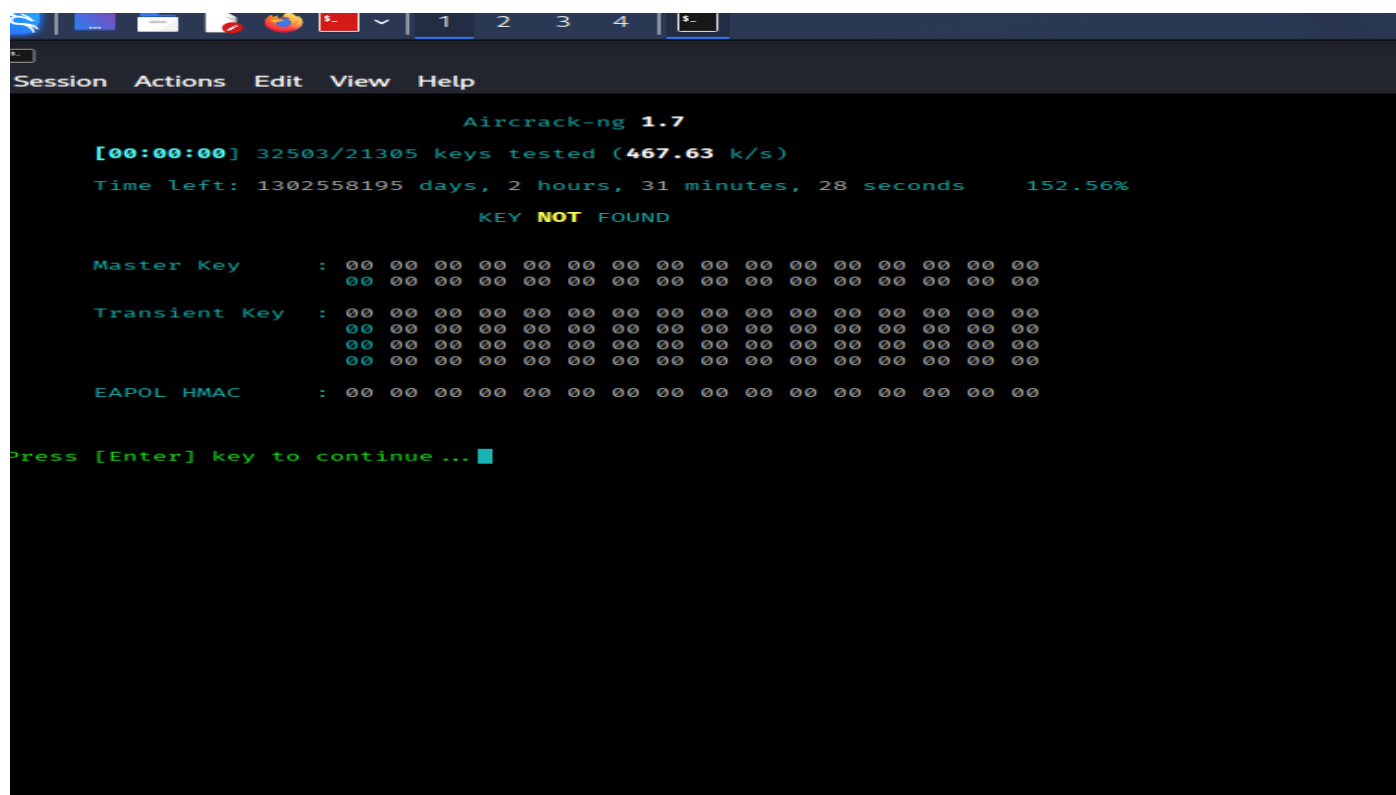
As a result, no credential interception activity occurred during the assessment.

Security Impact

Organizations without wireless intrusion detection controls remain vulnerable to rogue access point impersonation attacks.

Offline Password Attack Assessment

Screenshot – Offline Password Attack Results



Filename: 05.6-offline-capture.png

Description

The captured WPA2 handshake was subjected to an offline dictionary attack assessment.

The objective was to determine whether weak wireless credentials could be recovered using a predefined wordlist.

Observation

No password was recovered during testing.

The tool reported:

No Key Found

Security Impact

The assessment suggests that the wireless passphrase was not vulnerable to the wordlist used during testing.

However, successful handshake capture still demonstrates exposure to offline password attack attempts if weaker passwords are utilized in the future.

12. Wireshark Wireless Traffic Analysis

Wireless traffic observation was performed using Wireshark to examine 802.11 wireless communications associated with the target wireless network.

The objective of this phase was to analyze wireless management frames, beacon transmissions, client communications, authentication activities, and network behavior visible within the wireless environment.

Traffic analysis assists security professionals in understanding wireless network operations, identifying potential security weaknesses, validating encryption implementations, and detecting suspicious wireless activities.

Parameter Value

SSID BET9A@IKAY SHOP

BSSID B4:0F:3B:2C:8B:B8

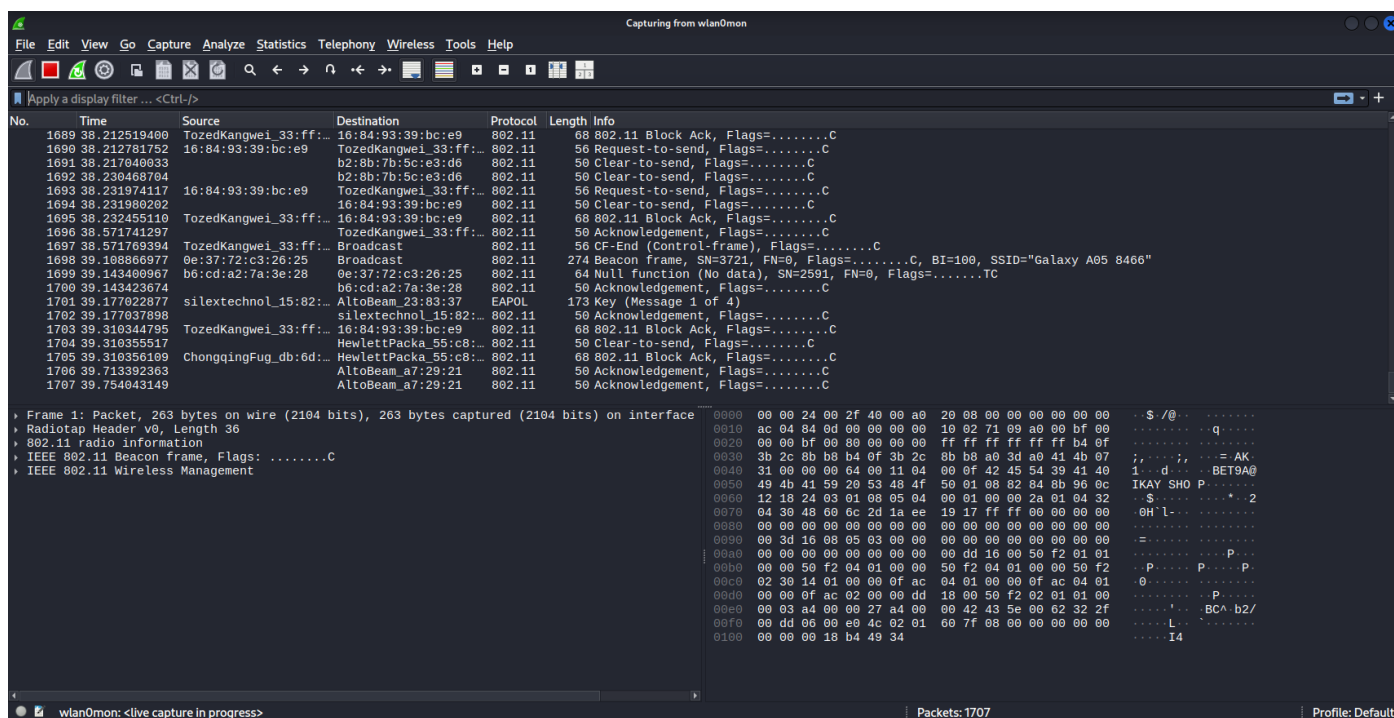
Channel 8

Security WPA2-PSK

Cipher CCMP

Live Wireless Packet Capture

Screenshot – Live Packet Capture



Filename: 06-live-packet-capture.png

Description

The screenshot shows a live packet capture session conducted using Wireshark while monitoring wireless traffic through the wireless adapter operating in monitor mode.

Several IEEE 802.11 wireless management and control frames were observed during the capture process, including:

- Beacon Frames
- Block Acknowledgement Frames
- Request-To-Send (RTS)
- Clear-To-Send (CTS)
- EAPOL Authentication Frames

The capture confirms active wireless communications occurring between wireless clients and nearby access points.

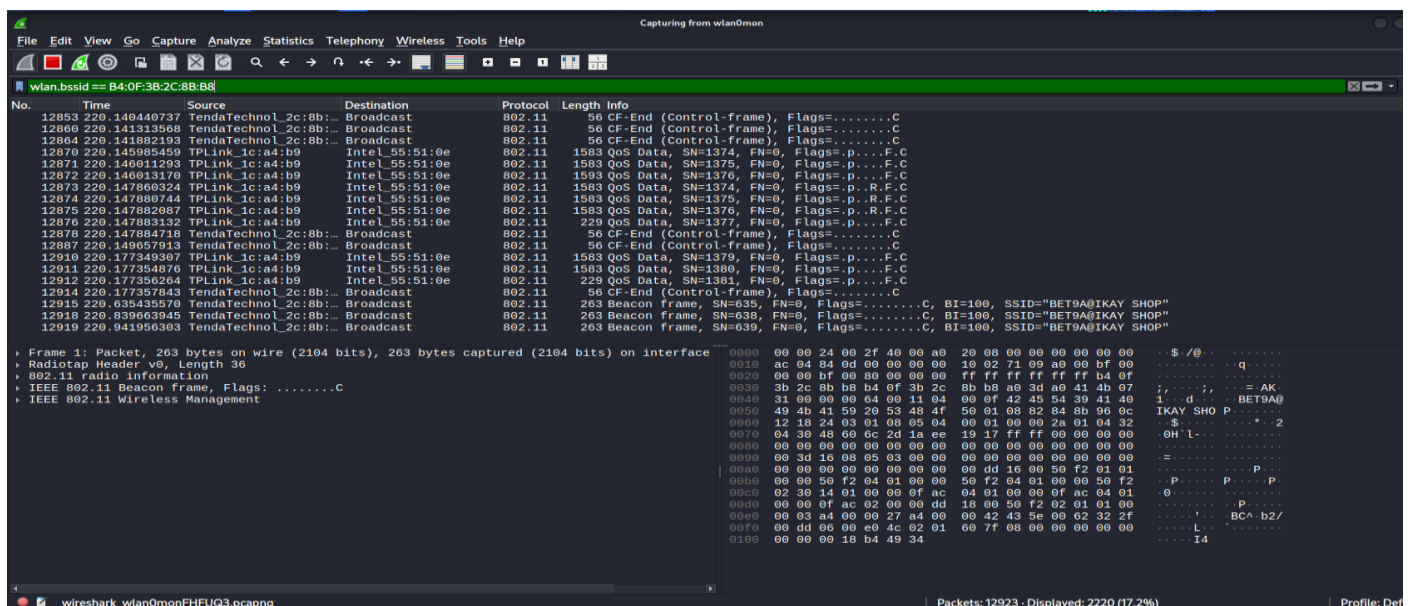
Security Relevance

The presence of EAPOL frames demonstrates wireless authentication exchanges used during WPA/WPA2 connection establishment.

Wireless management traffic visibility confirms that the monitoring setup successfully captured wireless communications required for analysis and security assessment activities.

Target Network Traffic Identification

Screenshot – Target Network Traffic Focus



Filename: 06.1-target-focus.png

Description

The screenshot shows Wireshark traffic filtered specifically for the target wireless network using the access point BSSID.

Filter applied:

wlan.bssid == B4:0F:3B:2C:8B:B8

The filtered capture displays only traffic associated with the target access point.

Observed information includes:

- Beacon Frames
- QoS Data Frames
- Control Frames
- Broadcast Traffic
- Client Communications

The SSID **BET9A@IKAY SHOP** is visible within beacon frame transmissions.

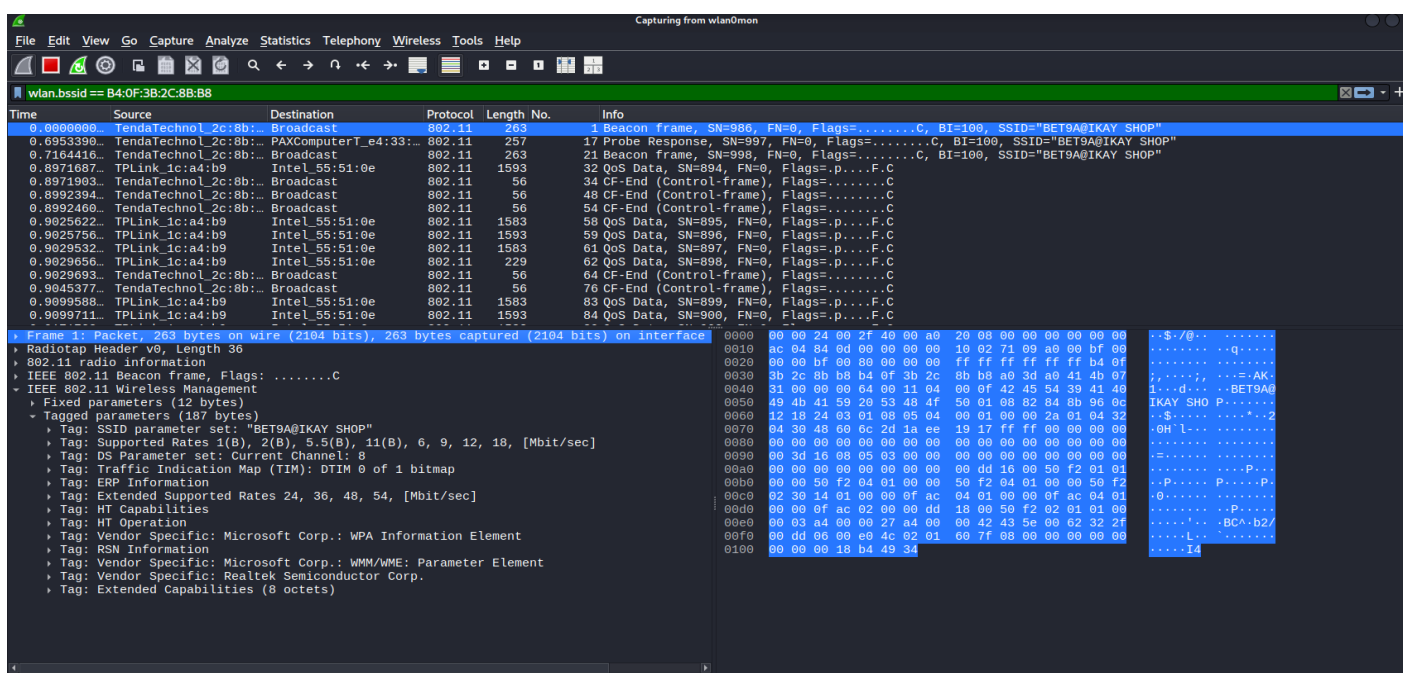
Security Relevance

Filtering traffic by BSSID enables focused analysis of a specific wireless network and reduces noise generated by surrounding wireless devices.

This process assists security analysts in examining network behavior, identifying active clients, validating wireless configurations, and monitoring communication patterns associated with the target infrastructure.

Packet Detail Analysis

Screenshot – Packet Detail Analysis



Filename: 06.2-packet-detail-analysis.png

Description

A Beacon Frame transmitted by the target access point (**BET9A@IKAY SHOP**) was selected and analyzed using Wireshark.

The packet details pane was expanded to inspect the wireless management frame and its associated tagged parameters. This analysis provided detailed information regarding the wireless network configuration and security settings.

Findings

The Beacon Frame analysis revealed the following information:

Parameter	Value
SSID	BET9A@IKAY SHOP
BSSID	B4:0F:3B:2C:8B:B8
Channel	8
Wireless Standard	IEEE 802.11
Security Protocol	WPA2
Authentication Method	PSK (Pre-Shared Key)
Encryption Cipher	CCMP (AES)
Vendor Information	Tenda Technology
HT Capabilities	Enabled

The tagged parameter section also revealed:

- Supported wireless data rates
- Current operating channel
- RSN (Robust Security Network) information
- WPA security information elements
- WMM/WME quality of service capabilities
- Extended wireless capabilities

Observation

The target network was configured with WPA2-PSK using AES (CCMP) encryption.

Although WPA2 remains significantly more secure than WEP and WPA, the use of a pre-shared key introduces exposure to:

- Handshake capture attacks
- Offline password cracking attempts
- Weak password exploitation
- PMKID-based attacks

Security Significance

Beacon Frames continuously broadcast information about the wireless network and can be collected passively by attackers without authenticating to the network.

Information obtained from Beacon Frames can support:

- Wireless reconnaissance
- Target identification
- Channel discovery
- Encryption analysis
- Wireless attack preparation

Protocol Hierarchy Statistics

Screenshot – Protocol Hierarchy Statistics

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU's
Frame	100.0	9803	100.0	5112463	43 k	0	0	0	9803
IEEE 802.11 Radiotap Capture header	100.0	9803	6.7	343760	2,936	0	0	0	9803
802.11 radio information	100.0	9803	0.0	0	0	0	0	0	9803
IEEE 802.11 wireless LAN	100.0	9803	5.3	272314	2,325	3910	72688	620	9803
Logical-Link Control	0.0	2	0.0	16	0	0	0	0	2
802.1X Authentication	0.0	2	0.0	310	2	2	310	2	2
Data	60.1	5891	82.7	4228462	36 k	5891	4228462	36 k	5891

Filename: 06.3-traffic-statistics.png

Description

Wireshark's Protocol Hierarchy Statistics feature was used to analyze the overall composition of captured wireless traffic.

This analysis provides visibility into the types of packets being transmitted within the wireless environment and helps identify authentication activity, management traffic, and user-generated communications.

Findings

The capture contained approximately:

Traffic Type	Packets
Total Frames	9,803
IEEE 802.11 Wireless LAN	9,803
Data Frames	5,891
Authentication Frames	2

Observation

The majority of captured traffic consisted of wireless data frames representing normal client-to-access point communications.

The statistics also revealed:

- Active wireless communications between clients and the access point
- Authentication-related traffic
- Continuous wireless management activity
- Significant user-generated wireless data traffic

Data frames represented approximately **60.1%** of all captured packets, indicating active utilization of the wireless network during the assessment period.

Security Significance

The presence of authentication traffic confirms that wireless authentication events occurred during the capture period.

From an attacker perspective, periods of active wireless authentication increase opportunities for:

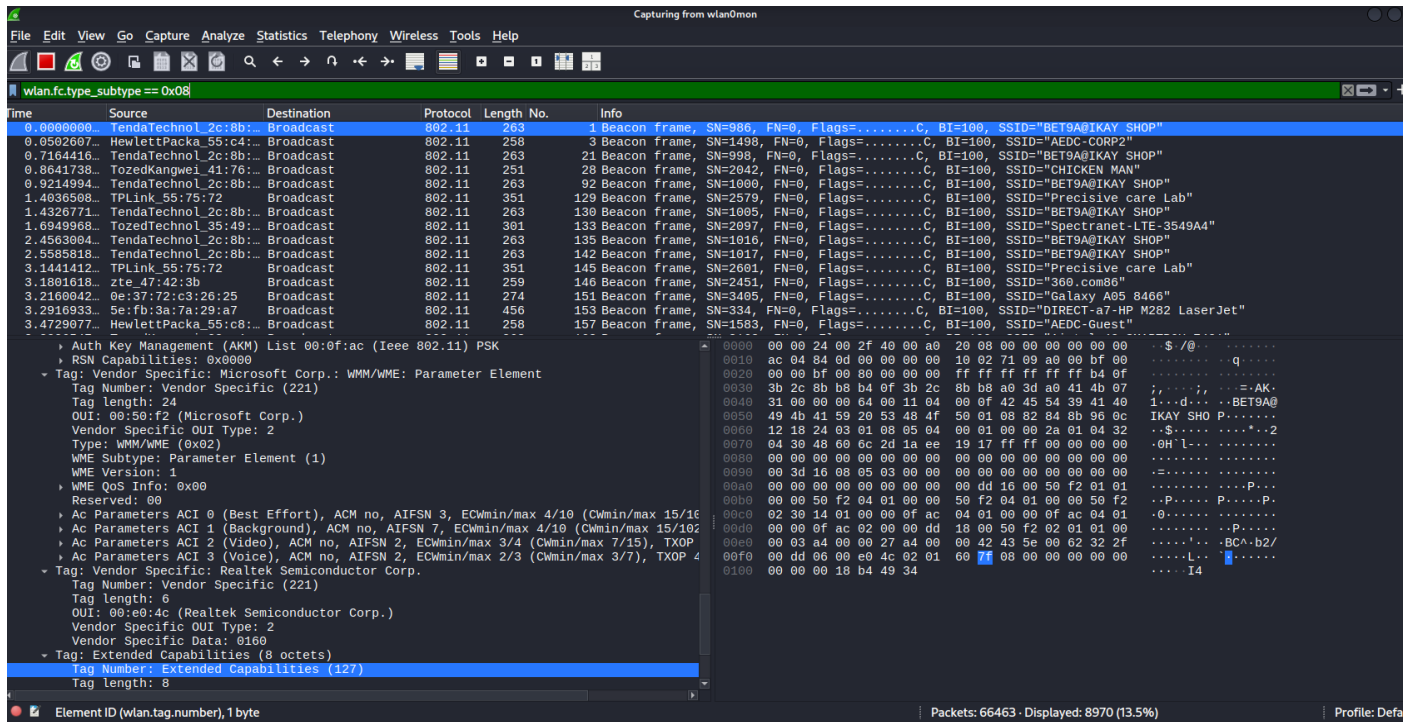
- WPA/WPA2 handshake capture
- Credential interception attempts
- Network reconnaissance
- Wireless traffic analysis

From a defender perspective, protocol hierarchy statistics provide valuable insight into:

- Wireless network utilization
- Authentication behavior
- Abnormal traffic patterns
- Potential wireless security incidents

Wireless Traffic Observation

Screenshot – Wireless Traffic Observation



Filename: 06.4-wireless-traffic-observation.png

Description

The screenshot shows detailed inspection of wireless beacon frames transmitted by the target access point.

Beacon frames provide important information about wireless network configuration and operational parameters.

The analysis revealed:

Parameter	Observation
SSID	BET9A@IKAY SHOP
Authentication	WPA2-PSK
Encryption	CCMP (AES)
Channel	8
Vendor Information	Realtek / Microsoft Vendor Elements
Extended Capabilities Present	
WMM/QoS Support	Enabled

The beacon frame continuously advertises network information to nearby wireless devices seeking network connectivity.

Security Relevance

Beacon frame analysis allows defenders and security assessors to identify:

- Wireless encryption mechanisms
- Authentication methods
- Supported wireless capabilities
- Access point configuration details
- Vendor-specific implementation information

This information assists in evaluating the overall security posture of the wireless infrastructure and identifying possible weaknesses associated with wireless network deployment.

Assessment Observation

Analysis of captured wireless traffic showed that the target network was actively transmitting management, control, and data frames consistent with normal wireless operations.

No evidence of WPA3 deployment was observed during traffic inspection. The network relied on WPA2-PSK authentication, which remains widely used but is susceptible to password-based attacks when weak credentials are deployed.

The observed wireless traffic supported findings identified during reconnaissance, handshake capture testing, and wireless security assessment activities conducted throughout this project.

13. WPA/WPA2/WPA3 Security Analysis

Wireless networks commonly use WPA, WPA2, or WPA3 to protect communications and restrict unauthorized access.

WPA is considered obsolete and should no longer be used in enterprise environments due to known cryptographic weaknesses. WPA2 remains the most widely deployed wireless security protocol and utilizes AES encryption to protect wireless communications. WPA3 is the latest wireless security standard and provides enhanced protection against password-based attacks through improved authentication mechanisms.

During this assessment, the target network BET9A@IKAY SHOP was identified as operating with WPA2-Personal using AES encryption.

Identified Wireless Security Configuration

Wireless reconnaissance activities revealed that the target network operated using WPA2-Personal security with AES encryption.

Wireless Security Configuration

Parameter	Observed Value
SSID	BET9A@IKAY SHOP
Security Protocol	WPA2-Personal (WPA2-PSK)
Cipher Suite	CCMP (AES)
Frequency Band	2.4 GHz
Channel	8
Authentication Method	Pre-Shared Key (PSK)

The use of WPA2 with AES encryption provides significantly stronger protection than legacy WEP and WPA implementations. However, the security of WPA2-Personal environments remains highly dependent on the strength and complexity of the configured wireless password.

WPA Security Analysis

Wi-Fi Protected Access (WPA) was introduced as a temporary replacement for the insecure WEP protocol.

WPA utilizes the Temporal Key Integrity Protocol (TKIP) to improve wireless security. Although stronger than WEP, WPA contains several known weaknesses and is no longer considered secure for enterprise environments.

Key Weaknesses of WPA

- Uses the outdated TKIP encryption mechanism.

- Vulnerable to packet injection attacks.
- Susceptible to cryptographic weaknesses.
- Does not provide adequate protection against modern wireless threats.

Enterprise Security Assessment

WPA should not be used within modern enterprise environments and should be disabled wherever possible.

WPA2 Security Analysis

WPA2 remains one of the most widely deployed wireless security standards globally and provides substantially stronger protection through the use of Advanced Encryption Standard (AES) encryption.

The target network was observed utilizing WPA2 with CCMP (AES), which is currently considered secure when combined with strong authentication credentials.

Security Strengths

- Strong AES-based encryption.
- Industry-standard wireless security protocol.
- Resistant to many attacks affecting older wireless technologies.
- Widely supported across enterprise infrastructure.

Security Weaknesses

Despite its strengths, WPA2-Personal remains vulnerable to several attack techniques when weak passwords are used.

Potential attack vectors include:

- Handshake capture attacks
- PMKID attacks
- Offline password cracking
- Dictionary attacks
- Brute-force attacks against weak passphrases

Assessment Finding

The target network's WPA2 implementation provides an acceptable level of security; however, overall protection remains dependent on password complexity and wireless security management practices.

WPA3 Security Analysis

WPA3 is the latest wireless security standard developed by the Wi-Fi Alliance and provides enhanced protection against password-based attacks.

WPA3 introduces Simultaneous Authentication of Equals (SAE), commonly referred to as the Dragonfly Handshake, which significantly improves resistance against offline password attacks.

Security Advantages of WPA3

- Improved protection against password guessing attacks.
- Enhanced resistance to offline cracking attempts.
- Forward secrecy protections.
- Stronger cryptographic implementation.
- Improved protection for public and enterprise wireless environments.

Enterprise Security Benefits

Organizations adopting WPA3 benefit from stronger authentication controls and reduced exposure to credential theft and wireless compromise attempts.

Comparative Security Analysis

Feature	WPA	WPA2	WPA3
Encryption	TKIP	AES-CCMP	AES-CCMP + SAE
Security Level	Low	High	Very High
Offline Password Attack Resistance	Weak	Moderate	Strong
Enterprise Suitability	Not Recommended	Acceptable	Recommended
Current Industry Status	Deprecated	Widely Used	Preferred Standard

Wireless Security Risks Identified

Based on the assessment findings, the following wireless security risks were identified:

Risk 1: Password-Based Authentication

The use of WPA2-Personal relies entirely on the strength of a shared password.

Weak passwords increase the likelihood of successful offline password cracking attempts following handshake capture.

Risk 2: Absence of WPA3 Protections

The target network does not currently benefit from WPA3 security enhancements, leaving it exposed to attack techniques that WPA3 was specifically designed to mitigate.

Risk 3: Credential Exposure Risk

Wireless credentials may become vulnerable if users select weak passwords or reuse passwords across multiple systems.

Risk 4: Legacy Wireless Compatibility

Organizations maintaining support for older wireless standards may unintentionally increase their attack surface.

Security Recommendations

To strengthen wireless security and reduce enterprise risk exposure, the following recommendations are provided:

1. Migrate to WPA3

Upgrade wireless infrastructure to WPA3 wherever supported by hardware and client devices.

2. Enforce Strong Passphrases

Implement password policies requiring:

- Minimum 16-character length
- Uppercase and lowercase characters
- Numbers
- Special characters

3. Implement Enterprise Authentication

Replace WPA2-Personal with WPA2/WPA3 Enterprise utilizing 802.1X authentication and centralized identity management.

4. Disable Legacy Protocols

Disable WEP, WPA, and TKIP support across all wireless infrastructure.

5. Conduct Periodic Wireless Assessments

Perform routine wireless security assessments to identify misconfigurations, weak credentials, and emerging wireless threats.

Conclusion

The wireless assessment identified WPA2-Personal with AES encryption as the security mechanism protecting the target network BET9A@IKAY SHOP. While WPA2 remains a secure and widely accepted wireless standard, its effectiveness is highly dependent on password strength and proper wireless security management.

The absence of WPA3 protections introduces additional risk exposure, particularly against modern password-based attack techniques. Organizations should consider migrating to WPA3 and implementing enterprise-grade authentication controls to improve wireless security resilience and reduce the likelihood of unauthorized wireless access.

14. Rogue Device Detection Report

Rogue wireless devices represent one of the most significant threats to enterprise wireless environments. Unauthorized access points, wireless repeaters, personal hotspots, and malicious wireless infrastructure can provide attackers with opportunities to bypass security controls, intercept communications, and gain unauthorized access to corporate resources.

As part of this assessment, wireless reconnaissance activities were performed to identify unauthorized wireless infrastructure operating within the assessment area.

Rogue Device Detection Methodology

The following activities were performed during rogue device detection:

Wireless Network Discovery

Wireless reconnaissance was conducted using:

- Airodump-ng
- Airededdon
- Wireshark

The objective was to identify:

- Unauthorized access points
- Duplicate SSIDs
- Suspicious BSSIDs
- Evil Twin infrastructure
- Open wireless networks
- Unexpected wireless devices
- Guest wireless exposure
- Wireless client activity

Wireless Network Enumeration

Detected wireless networks were analyzed to determine:

- SSID names
- BSSID addresses
- Encryption methods
- Signal strength
- Channel allocations
- Client associations

This information was used to distinguish legitimate wireless infrastructure from potentially unauthorized devices.

Assessment Findings

Authorized Wireless Network Identified

The primary wireless network selected for assessment was:

Parameter Value

SSID BET9A@IKAY SHOP

BSSID B4:0F:3B:2C:8B:B8

Security WPA2-PSK

Channel 8

Cipher CCMP (AES)

The access point consistently appeared throughout the assessment and maintained active client associations.

Duplicate SSID Analysis

Wireless reconnaissance did not reveal any duplicate instances of:

BET9A@IKAY SHOP

No additional access points were observed broadcasting the same SSID using different BSSID addresses.

Assessment Result

No evidence of an active Evil Twin access point was identified during the assessment period.

Unauthorized Access Point Detection

All detected wireless networks appeared to be independent wireless networks operating within the surrounding environment.

No access point was positively identified as:

- Rogue infrastructure
- Unauthorized enterprise access point
- Malicious hotspot
- Unauthorized repeater
- Unauthorized bridge device

Assessment Result

No confirmed rogue access point was detected during testing.

Potential Rogue Device Risks

Although no rogue device was confirmed during the assessment, enterprise environments remain exposed to several wireless threats.

Examples include:

Rogue Access Points

Employees may connect unauthorized wireless routers to internal networks.

Potential impact:

- Security control bypass
- Network exposure
- Unauthorized access

Evil Twin Access Points

Attackers may create fake wireless networks that impersonate legitimate enterprise SSIDs.

Potential impact:

- Credential theft
- Traffic interception
- Session hijacking

Personal Mobile Hotspots

Users may unintentionally create unauthorized wireless networks through mobile devices.

Potential impact:

- Data leakage
- Security policy violations
- Shadow IT risks

Wireless Repeaters and Extenders

Unauthorized wireless extenders may weaken network visibility and security monitoring.

Potential impact:

- Expanded attack surface
- Reduced security control effectiveness

Risk Assessment

Risk Category	Assessment
Confirmed Rogue AP	Not Detected
Evil Twin AP	Not Detected
Duplicate SSID	Not Detected
Unauthorized Wireless Bridge	Not Detected
Personal Hotspot Activity	Not Observed
Potential Exposure	Moderate

Recommendations

The following controls are recommended to reduce rogue device risks:

Continuous Wireless Monitoring

Deploy wireless monitoring solutions capable of identifying unauthorized wireless infrastructure.

Wireless Intrusion Detection Systems (WIDS)

Implement WIDS technologies to automatically detect rogue access points and suspicious wireless behavior.

Wireless Intrusion Prevention Systems (WIPS)

Deploy WIPS controls capable of actively responding to rogue wireless devices.

Periodic Wireless Audits

Conduct routine wireless assessments to identify unauthorized infrastructure.

Network Access Control (NAC)

Implement NAC solutions to prevent unauthorized devices from joining enterprise networks.

Security Awareness Training

Educate employees regarding the risks associated with personal hotspots, wireless extenders, and unauthorized networking devices.

Conclusion

No confirmed rogue access point was identified during the assessment of the BET9A@IKAY SHOP wireless environment. However, the assessment demonstrated the importance of continuous wireless monitoring and rogue device detection capabilities within enterprise environments.

Organizations should maintain proactive wireless monitoring programs to identify unauthorized infrastructure before it can be leveraged by attackers for credential theft, network compromise, or unauthorized access.

15. Wireless Governance Failure Analysis

Wireless security incidents are rarely caused solely by technology failures. In many organizations, security breaches occur due to weaknesses in governance, policy enforcement, operational oversight, user awareness, and security monitoring.

During the wireless assessment of the target network **BET9A@IKAY SHOP**, several governance-related weaknesses were identified that could increase the organization's exposure to unauthorized wireless access, credential compromise, rogue device connections, and potential data breaches.

The purpose of this section is to analyze organizational and operational weaknesses that may contribute to wireless security incidents and compliance failures.

Governance Failure 1: Weak Wireless Authentication Management

Observation

The target network utilized WPA2-Personal authentication based on a shared wireless password.

Risk

Shared passwords create accountability challenges because multiple users may possess the same credential.

If the password is disclosed, shared externally, or reused across multiple systems, unauthorized access may occur without easy attribution to a specific user.

Potential Impact

- Unauthorized network access
- Credential leakage
- Difficulty tracking user activity
- Increased insider threat risk

Recommendation

Implement WPA2/WPA3 Enterprise authentication using 802.1X and centralized identity management.

Governance Failure 2: Lack of Wireless Access Control Policies

Observation

No evidence was available indicating the existence of formal wireless access control policies governing:

- Employee devices
- BYOD devices
- Guest users
- Third-party contractors

Risk

Uncontrolled device access may increase exposure to malware infections, rogue devices, and unauthorized wireless connections.

Potential Impact

- Unauthorized device access
- Increased attack surface
- Lateral movement opportunities

Recommendation

Establish formal wireless access control policies and device enrollment procedures.

Governance Failure 3: Inadequate Wireless Monitoring and Detection

Observation

The assessment identified that wireless attacks such as:

- Deauthentication attacks
- Handshake capture attempts
- Rogue access point deployment
- Credential interception attacks

could potentially occur without dedicated wireless monitoring controls.

Risk

Attackers may operate undetected within the wireless environment.

Potential Impact

- Delayed incident detection
- Unauthorized wireless activity
- Increased dwell time for attackers

Recommendation

Deploy Wireless Intrusion Detection Systems (WIDS) and Wireless Intrusion Prevention Systems (WIPS).

Governance Failure 4: Absence of Rogue Access Point Detection Procedures

Observation

No formal rogue access point detection process was observed during the assessment.

Risk

Attackers may deploy unauthorized access points that imitate legitimate wireless networks.

Potential Impact

- Credential theft
- Evil Twin attacks
- Adversary-in-the-Middle attacks
- Unauthorized network access

Recommendation

Implement continuous rogue AP monitoring and periodic wireless audits.

Governance Failure 5: Weak Wireless Security Awareness

Observation

Users may unknowingly connect to malicious wireless networks that closely resemble legitimate organizational SSIDs.

Risk

Attackers can exploit human behavior through:

- Evil Twin attacks
- Fake captive portals
- Credential harvesting attacks

Potential Impact

- User credential compromise
- Unauthorized access
- Sensitive data exposure

Recommendation

Conduct regular wireless security awareness training.

Governance Failure 6: Insufficient Network Segmentation

Observation

Enterprise environments often fail to properly separate:

- Guest Wi-Fi
- Administrative systems

- CCTV infrastructure
- IoT devices
- BYOD networks

Risk

A compromise of one wireless segment may provide attackers with access to sensitive internal resources.

Potential Impact

- Lateral movement
- Unauthorized access to business systems
- Increased attack propagation

Recommendation

Implement VLAN-based segmentation and Zero Trust network architecture principles.

Governance Failure 7: Failure to Adopt Modern Wireless Security Standards

Observation

The target environment currently relies on WPA2-Personal authentication.

Risk

Although WPA2 remains widely used, it lacks several protections available in WPA3.

Potential Impact

- Increased susceptibility to password-based attacks
- Reduced resistance to credential compromise

Recommendation

Develop a phased migration strategy toward WPA3 deployment.

Compliance and Regulatory Risks

The identified governance weaknesses may contribute to violations of industry security requirements and regulatory obligations.

Potential compliance concerns include:

Compliance Area	Potential Risk
Access Control	Unauthorized wireless access
Data Protection	Credential interception
Monitoring & Logging	Lack of wireless visibility

Compliance Area	Potential Risk
Asset Management	Rogue device connections
Security Awareness	User susceptibility to attacks
Network Security	Weak segmentation controls

Failure to address these issues may increase the likelihood of data breaches, regulatory findings, and operational disruptions.

Conclusion

The assessment identified several governance and operational weaknesses that may increase exposure to wireless security threats. While technical controls such as WPA2 encryption provide a baseline level of protection, effective wireless security requires strong governance, monitoring, access control, user awareness, segmentation, and continuous security oversight.

Addressing these governance failures will significantly improve the organization's ability to prevent unauthorized wireless access, detect malicious activity, and maintain a resilient enterprise wireless security posture.

16. MITRE ATT&CK Mapping

The MITRE ATT&CK Framework provides a globally recognized knowledge base of adversary tactics and techniques based on real-world attack behavior.

During this wireless security assessment, several activities performed during reconnaissance, wireless monitoring, handshake capture analysis, traffic inspection, and password attack simulation were mapped against relevant MITRE ATT&CK techniques to demonstrate how threat actors may target wireless environments.

The mapping helps identify attacker objectives, attack paths, and defensive controls that should be implemented to improve enterprise wireless security.

MITRE ATT&CK Mapping Summary

MITRE ID	Technique	ATT&CK Tactic	Assessment Relevance
T1595	Active Scanning	Reconnaissance	Wireless network discovery
T1669	Wi-Fi Networks	Initial Access	Wireless network targeting and access attempts
T1040	Network Sniffing	Credential Access, Discovery	Wireless packet capture using Wireshark
T1110.002	Password Cracking	Credential Access	Offline WPA2 password attack assessment
T1557	Adversary-in-the-Middle	Credential Access, Collection	Evil Twin attack assessment

1. Technique Analysis – T1595 Active Scanning

MITRE Technique

Technique ID: T1595

Technique Name: Active Scanning

Tactic: Reconnaissance

Description

Adversaries perform active scanning to gather information about target infrastructure before attempting exploitation or unauthorized access.

Assessment Evidence

During the assessment, Airodump-ng was used to identify:

- Nearby wireless networks
- SSIDs
- BSSIDs
- Channel allocations
- Signal strengths
- Encryption types
- Connected clients

This activity mirrors how attackers conduct reconnaissance prior to targeting wireless infrastructure.

Potential Risk

Attackers can identify:

- Weakly secured wireless networks
- Legacy protocols
- High-value targets
- Network characteristics

Recommendation

Implement wireless monitoring and reduce unnecessary SSID broadcasting where operationally feasible.

2. Technique Analysis – T1669 Wi-Fi Networks

MITRE Technique

Technique ID: T1669

Technique Name: Wi-Fi Networks

Tactic: Initial Access

Description

Threat actors may gain initial access by connecting to wireless networks using valid credentials, compromised credentials, or insecure wireless configurations.

Assessment Evidence

The target network:

SSID: BET9A@IKAY SHOP

was identified and selected for controlled wireless assessment activities.

The assessment demonstrated how a threat actor could target wireless infrastructure as an initial access vector.

Potential Risk

Unauthorized users obtaining wireless access may:

- Reach internal resources
- Access sensitive systems
- Conduct further attacks

Recommendation

Implement WPA3 Enterprise and network segmentation controls.

3. Technique Analysis – T1040 Network Sniffing

MITRE Technique

Technique ID: T1040

Technique Name: Network Sniffing

Tactics:

- Credential Access
- Discovery

Description

Threat actors may capture network traffic to obtain credentials, discover network information, and identify valuable assets.

Assessment Evidence

Wireshark was used to capture and analyze wireless traffic.

Observed traffic included:

- Beacon Frames
- Probe Requests
- Authentication Frames
- Management Frames
- Wireless network advertisements

Potential Risk

Attackers may use packet captures to:

- Discover devices
- Identify wireless infrastructure
- Monitor communications

- Collect authentication information

Recommendation

Use strong encryption and deploy wireless intrusion detection systems.

4. Technique Analysis – T1110.002 Password Cracking

MITRE Technique

Technique ID: T1110.002

Technique Name: Password Cracking

Tactic: Credential Access

Description

Threat actors may attempt offline password cracking against captured authentication material.

Assessment Evidence

A WPA2 handshake was successfully captured during assessment activities.

A controlled offline password attack was performed against the captured handshake.

The attack did not recover the wireless password.

Assessment Result

No password was recovered from the tested wordlist.

Potential Risk

Weak wireless passwords could be successfully recovered through dictionary or brute-force attacks.

Recommendation

Enforce strong password policies and transition to WPA3 authentication.

5. Technique Analysis – T1557 Adversary-in-the-Middle

MITRE Technique

Technique ID: T1557

Technique Name: Adversary-in-the-Middle

Tactics

- Credential Access
- Collection

Description

Threat actors may position themselves between users and legitimate services to intercept credentials and communications.

Assessment Evidence

Airgeddon was used to review Evil Twin attack capabilities.

The assessment demonstrated how a rogue wireless access point could impersonate a legitimate network.

No client connected to the simulated Evil Twin environment during testing.

Potential Risk

Successful Evil Twin attacks may result in:

- Credential theft
- Session hijacking
- Traffic interception
- User compromise

Recommendation

Deploy rogue AP detection, user awareness training, and certificate validation controls.

Wireless Attack Chain Mapping

The following attack chain demonstrates how a real-world attacker could progress through the wireless environment.

Phase	MITRE Technique	Description
Reconnaissance	T1595	Discover wireless networks
Initial Access	T1669	Target Wi-Fi infrastructure
Credential Access	T1040	Monitor wireless traffic
Credential Access	T1110.002	Attempt password cracking
Collection	T1557	Conduct Evil Twin attack
Impact	Network Compromise	Unauthorized wireless access

Conclusion

The MITRE ATT&CK mapping demonstrates that the activities observed during this wireless assessment closely align with real-world adversary techniques used against enterprise wireless environments.

The identified techniques highlight potential attack paths that could be leveraged to gain unauthorized access, intercept credentials, monitor wireless communications, and compromise enterprise resources.

Implementing the recommendations provided throughout this report will significantly reduce exposure to these attack techniques and strengthen the organization's wireless security posture.

17. Wireless Risk Register

A wireless risk register was developed to document the security risks identified during the wireless infrastructure assessment. The purpose of the register is to prioritize remediation activities and provide management with a clear understanding of the risks affecting the organization's wireless environment.

Risk ratings were determined based on the likelihood of exploitation and the potential impact to confidentiality, integrity, and availability.

Wireless Risk Register

Risk ID	Finding	Likelihood	Impact	Risk Rating	MITRE Mapping	Recommendation
WR-001	WPA2 Handshake Capture Exposure	Medium	High	High	T1110.002	Enforce strong passwords and migrate to WPA3
WR-002	Offline Password Cracking Risk	Medium	High	High	T1110.002	Implement complex passphrases and password rotation
WR-003	Wireless Network Discovery Exposure	High	Medium	High	T1595	Continuous wireless monitoring and periodic assessments
WR-004	Evil Twin Attack Exposure	Medium	High	High	T1557	Deploy rogue AP detection and user awareness training
WR-005	Wireless Traffic Monitoring Exposure	Medium	Medium	Medium	T1040	Encrypt traffic and implement WIDS/WIPS
WR-006	Weak Wireless Authentication Management	Medium	High	High	T1669	Deploy WPA2/WPA3 Enterprise authentication
WR-007	Lack of Wireless Segmentation	High	High	Critical	T1669	Implement VLAN segmentation and Zero Trust controls
WR-008	Unauthorized Device	Medium	High	High	T1669	Implement Network Access Control (NAC)

Risk ID	Finding	Likelihood	Impact	Risk Rating	MITRE Mapping	Recommendation
	Connection Risk					
WR-009	Absence of Rogue Access Point Monitoring	Medium	High	High	T1557	Conduct regular rogue AP detection scans
WR-010	Legacy Wireless Protocol Exposure	Low	High	Medium	T1669	Disable deprecated wireless protocols

Critical Risks Identified

The assessment identified several risks requiring immediate management attention.

WR-007 – Lack of Wireless Segmentation

This represents the most significant risk identified during the assessment.

Without proper segmentation between:

- Administrative networks
- Guest networks
- BYOD devices
- IoT infrastructure
- CCTV systems

an attacker gaining wireless access may be able to move between network segments and access sensitive resources.

Risk Rating: Critical

WR-001 – WPA2 Handshake Capture Exposure

The assessment successfully captured a WPA2 handshake from the target network.

Although the offline password attack was unsuccessful, weak passwords could allow attackers to recover credentials through dictionary or brute-force attacks.

Risk Rating: High

WR-004 – Evil Twin Attack Exposure

The assessment demonstrated the potential for attackers to deploy rogue wireless access points impersonating legitimate enterprise networks.

Successful attacks could lead to credential theft and unauthorized access.

Risk Rating: High

Risk Prioritization Matrix

Risk Rating	Priority Level	Action Required
Critical	Immediate	Remediate within 30 days
High	High Priority	Remediate within 60 days
Medium	Moderate Priority	Remediate within 90 days
Low	Routine Monitoring	Review periodically

Conclusion

The wireless risk assessment identified multiple risks that may increase the likelihood of unauthorized wireless access, credential compromise, rogue device connections, and network exposure.

The most significant risks relate to wireless authentication management, network segmentation, and wireless monitoring capabilities. Addressing these issues should be prioritized to improve the overall security posture of the wireless infrastructure.

18. Enterprise Wireless Redesign Architecture

Following the wireless security assessment conducted against the target wireless environment (BET9A@IKAY SHOP), several security weaknesses and operational risks were identified, including:

- WPA2 handshake capture exposure
- Evil Twin attack exposure
- Wireless reconnaissance visibility
- Weak authentication management
- Inadequate wireless segmentation
- Potential rogue access point risks
- Insufficient wireless monitoring capabilities

To address these issues, a redesigned enterprise wireless architecture is proposed to improve security, visibility, segmentation, monitoring, and access control across the organization.

The proposed architecture follows modern enterprise wireless security principles, including Zero Trust, Network Segmentation, Centralized Authentication, and Continuous Monitoring.

Current-State Security Challenges

The assessment identified the following weaknesses within the wireless environment:

Finding	Security Impact
WPA2-Personal Authentication	Shared password exposure
Handshake Capture Possible	Offline password attack risk
Evil Twin Attack Exposure	Credential theft risk
Weak Guest Isolation	Unauthorized access risk
Lack of NAC Controls	Untrusted device connections
Limited Wireless Monitoring	Delayed attack detection
Insufficient Segmentation	Lateral movement opportunities

These weaknesses increase the organization's exposure to unauthorized wireless access and credential compromise.

Proposed Enterprise Wireless Architecture

The redesigned wireless infrastructure should separate users, systems, and devices into dedicated security zones.

Proposed Wireless Segmentation

Wireless Zone	Purpose
Corporate Wi-Fi	Employee access
Guest Wi-Fi	Visitor internet access
BYOD Network	Employee personal devices
IoT Network	Smart devices and sensors
CCTV Network	Surveillance systems
Administrative Network	Critical business operations
Security Operations Network	Monitoring and management systems

Each wireless segment should operate independently using dedicated VLANs and access control policies.

Proposed Network Segmentation Model

VLAN	Network Purpose
VLAN 10	Corporate Users
VLAN 20	Guest Wi-Fi
VLAN 30	BYOD Devices
VLAN 40	IoT Devices
VLAN 50	CCTV Infrastructure
VLAN 60	Administrative Systems
VLAN 70	Security Monitoring Systems

Inter-VLAN communication should be restricted through firewall policies and access control rules.

Secure Wireless Authentication Design

The current WPA2-Personal model should be replaced.

Recommended Authentication Model

WPA3-Enterprise

Integrated with:

- RADIUS Server
- Active Directory
- Multi-Factor Authentication (MFA)

Benefits include:

- Individual user authentication
- Centralized access management
- Improved audit logging
- Reduced credential sharing
- Stronger protection against password attacks

Network Access Control (NAC)

A Network Access Control solution should be implemented to verify devices before granting wireless access.

NAC Functions

- Device identification
- Device health validation
- User authentication
- Policy enforcement
- Rogue device blocking

Only approved devices should be allowed to access enterprise resources.

Rogue Access Point Protection

A Wireless Intrusion Detection and Prevention System (WIDS/WIPS) should be deployed.

Monitoring Capabilities

- Rogue AP detection
- Evil Twin detection
- Deauthentication attack detection
- Unauthorized client detection
- Wireless anomaly monitoring

This capability directly addresses risks observed during Airgeddon testing.

Enterprise Wireless Monitoring Architecture

Wireless activity should be continuously monitored through centralized logging and security analytics platforms.

Monitoring Sources

- Wireless Controllers
- Access Points
- RADIUS Servers
- NAC Systems
- Firewalls
- DHCP Servers
- DNS Servers

Security Monitoring Platform

All logs should be forwarded to a SIEM platform such as:

- Splunk
- Elastic SIEM
- Microsoft Sentinel
- IBM QRadar

Guest Wi-Fi Security Design

Guest users should never share the same wireless environment as internal corporate systems.

Guest Network Controls

- Dedicated VLAN
- Internet-only access
- Captive portal authentication
- Bandwidth limitations
- Device isolation
- Firewall restrictions

This prevents guest devices from reaching internal resources.

BYOD Security Design

Employee-owned devices present additional security risks.

Recommended Controls

- Separate BYOD VLAN
- NAC validation
- Device registration
- Security posture checks
- Limited access permissions

BYOD devices should not be permitted direct access to critical enterprise systems.

IoT and CCTV Security Design

IoT devices and surveillance systems are frequently targeted due to weak security configurations.

Recommended Controls

- Dedicated IoT VLAN
- Dedicated CCTV VLAN
- Restricted outbound communication
- Firewall enforcement
- Continuous monitoring

These devices should never share the same network as corporate users.

Secure Remote Wireless Access

Remote employees should access enterprise resources through secure VPN services.

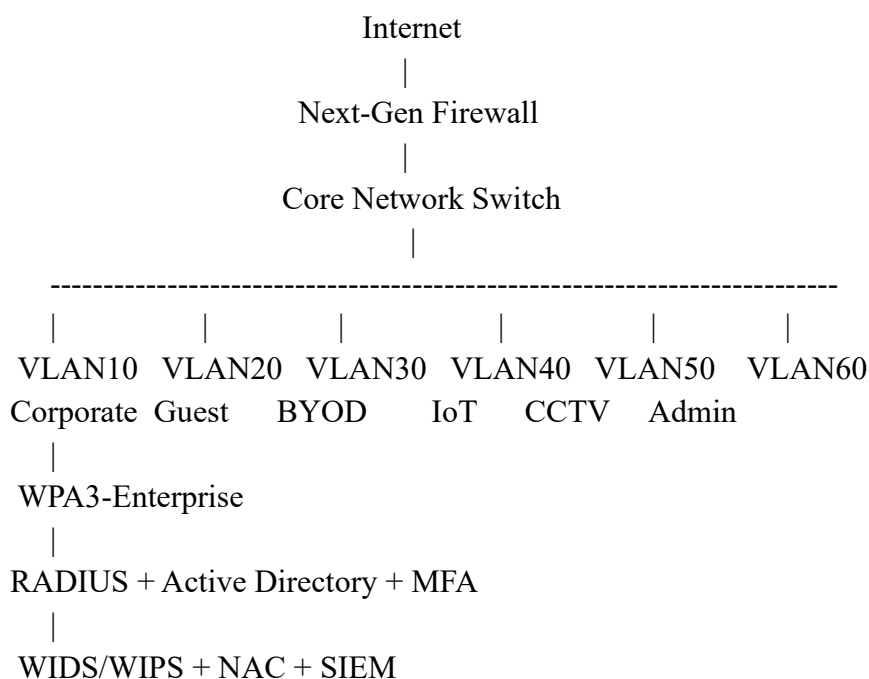
Security Controls

- VPN authentication
- Multi-Factor Authentication
- Device verification
- Session logging
- Conditional access policies

This reduces risks associated with remote wireless connections.

Proposed Enterprise Wireless Architecture

Suggested layout:



Benefits of the Proposed Design

Implementation of the redesigned architecture would provide:

- Stronger authentication
- Reduced password-sharing risks
- Improved wireless visibility
- Rogue device detection
- Enhanced monitoring capabilities
- Improved compliance readiness
- Better segmentation
- Reduced lateral movement opportunities
- Improved incident response capabilities

Conclusion

The proposed enterprise wireless architecture addresses the weaknesses identified during the assessment and aligns with modern enterprise security best practices. By implementing WPA3-Enterprise authentication, network segmentation, NAC, WIDS/WIPS, SIEM integration, and Zero Trust principles, KAPAS INTERNATIONAL can significantly reduce wireless security risks while improving operational visibility and security resilience.

19. Wireless SOC Monitoring Plan

Effective wireless security extends beyond prevention controls and requires continuous monitoring to detect malicious activities, unauthorized devices, rogue access points, credential attacks, and wireless intrusion attempts.

The purpose of the Wireless Security Operations Center (SOC) Monitoring Plan is to provide visibility into wireless activities across the enterprise environment and ensure that security incidents are identified and responded to in a timely manner.

The monitoring plan was developed based on the findings identified during the wireless assessment and attack simulation activities.

Wireless Security Monitoring Objectives

The objectives of the wireless SOC monitoring program include:

- Detect unauthorized wireless access attempts
- Detect rogue access points
- Detect Evil Twin attacks
- Monitor wireless authentication failures
- Identify wireless reconnaissance activities
- Detect wireless credential attacks
- Monitor deauthentication attacks
- Detect suspicious wireless clients
- Improve incident response capability
- Support forensic investigations

Log Sources for Monitoring

The SOC should collect and analyze logs from multiple wireless infrastructure components.

Log Source	Purpose
Wireless Access Points	Authentication and association events
Wireless Controllers	Centralized wireless management logs
RADIUS Server	User authentication logs

Log Source	Purpose
Network Access Control (NAC)	Device authorization events
DHCP Server	IP address assignment records
DNS Server	Name resolution activities
Firewall	Network access attempts
SIEM Platform	Correlation and alerting
WIDS/WIPS	Wireless intrusion detection events

Critical Wireless Events to Monitor

The following events should generate security alerts:

Unauthorized Wireless Connections

Indicators:

- Unknown device connections
- New MAC addresses
- Unapproved endpoints

Potential Impact:

- Unauthorized network access

MITRE Mapping:

- T1669 – Wi-Fi Networks

Wireless Reconnaissance Activity

Indicators:

- Excessive probe requests
- Wireless scanning behavior
- SSID enumeration

Potential Impact:

- Information gathering for future attacks

MITRE Mapping:

- T1595 – Active Scanning

Rogue Access Point Detection

Indicators:

- Duplicate SSIDs
- Unauthorized access points
- Suspicious BSSID activity

Potential Impact:

- Credential theft
- Evil Twin attacks

MITRE Mapping:

- T1557 – Adversary-in-the-Middle

Deauthentication Attack Detection

Indicators:

- Excessive deauthentication frames
- Client disconnections
- Authentication interruptions

Potential Impact:

- Denial of service
- Forced handshake capture

MITRE Mapping:

- T1557 – Adversary-in-the-Middle

Wireless Credential Attacks

Indicators:

- Multiple authentication failures
- Repeated connection attempts
- Password guessing behavior

Potential Impact:

- Credential compromise

MITRE Mapping:

- T1110.002 – Password Cracking

Security Use Cases

Use Case 1 – Rogue Access Point Detection

Objective:

Identify unauthorized wireless access points impersonating legitimate enterprise networks.

Alert Trigger:

- Duplicate SSID detected
- Unknown BSSID detected

Severity:

High

Use Case 2 – Wireless Reconnaissance Detection

Objective:

Identify wireless scanning and reconnaissance activities.

Alert Trigger:

- Excessive probe requests
- Unusual discovery activity

Severity:

Medium

Use Case 3 – Credential Attack Detection

Objective:

Identify password attacks against wireless infrastructure.

Alert Trigger:

- High authentication failure rates
- Multiple login attempts

Severity:

High

Use Case 4 – Deauthentication Attack Detection

Objective:

Detect attempts to disrupt wireless communications.

Alert Trigger:

- Large volumes of deauthentication frames

Severity:

High

Wireless Incident Response Workflow

When a wireless security incident is detected, the following response process should be followed:

Step 1 – Detection

SOC receives alert from:

- SIEM
- WIDS/WIPS
- Firewall
- Wireless Controller

Step 2 – Validation

Security analysts review:

- Logs
- Wireless events
- Device information
- User activity

Step 3 – Containment

Actions may include:

- Blocking MAC addresses
- Disabling wireless accounts
- Removing rogue access points
- Isolating affected devices

Step 4 – Investigation

Investigators collect:

- Packet captures
- Authentication logs
- Wireless controller logs
- Access point logs

Step 5 – Recovery

Restore normal operations and verify security controls.

Step 6 – Lessons Learned

Document findings and improve detection rules.

Recommended SOC Technologies

The following technologies are recommended:

Technology	Purpose
SIEM	Centralized monitoring
WIDS/WIPS	Wireless threat detection
NAC	Device access control
NGFW	Network protection
Endpoint Detection and Response (EDR)	Endpoint monitoring
Threat Intelligence Platform	Threat correlation

Conclusion

A dedicated wireless SOC monitoring capability is essential for detecting wireless threats before they impact enterprise operations. Continuous monitoring of wireless infrastructure, authentication systems, rogue access points, reconnaissance activity, and wireless attack indicators will significantly improve the organization's ability to detect and respond to security incidents.

20. Executive Briefing Report

This executive briefing summarizes the findings of the Comprehensive Wireless Infrastructure Security Assessment, Attack Simulation, and Remediation Analysis conducted against the wireless environment used as a representation of KAPAS INTERNATIONAL.

The assessment was performed to evaluate the effectiveness of existing wireless security controls, identify weaknesses that may expose the organization to unauthorized access, and provide recommendations for improving enterprise wireless security.

Wireless reconnaissance, attack simulation, traffic analysis, and security architecture reviews were conducted using industry-standard assessment tools and methodologies.

Assessment Summary

The assessment identified several wireless security risks that could potentially impact the confidentiality, integrity, and availability of enterprise systems.

The assessment demonstrated that wireless networks remain a significant attack surface for modern organizations and require continuous monitoring, strong authentication, effective segmentation, and proactive security controls.

The following assessment activities were performed:

- Wireless reconnaissance
- SSID identification
- BSSID identification
- Encryption analysis
- WPA2 handshake capture assessment
- Wireless traffic monitoring
- Deauthentication testing
- Evil Twin attack analysis
- Offline password attack assessment
- MITRE ATT&CK mapping
- Wireless risk assessment
- Enterprise wireless architecture review

Key Findings

Finding 1 – WPA2 Handshake Capture Exposure

The assessment successfully captured a WPA2 handshake from the target wireless network.

Although the offline password attack was unsuccessful, weak passwords could potentially allow attackers to recover wireless credentials through offline password-cracking techniques.

Business Risk:

- Unauthorized wireless access
- Credential compromise
- Network exposure

Risk Rating:

High

Finding 2 – Wireless Reconnaissance Exposure

The wireless infrastructure exposed information such as:

- SSID
- BSSID
- Channel allocation
- Encryption type

This information may assist attackers during the reconnaissance phase of an attack.

Business Risk:

- Increased attack surface visibility

Risk Rating:

Medium

Finding 3 – Evil Twin Attack Exposure

The assessment demonstrated that attackers may attempt to create rogue access points that imitate legitimate enterprise wireless networks.

Business Risk:

- Credential theft
- Unauthorized access
- User compromise

Risk Rating:

High

Finding 4 – Weak Network Segmentation

The assessment identified the need for stronger wireless network segmentation between:

- Corporate users
- Guest users
- BYOD devices
- IoT infrastructure
- CCTV systems

Business Risk:

- Lateral movement
- Unauthorized access
- Increased attack impact

Risk Rating:

Critical

MITRE ATT&CK Summary

The assessment findings were mapped to relevant MITRE ATT&CK techniques.

MITRE ID Technique

T1595	Active Scanning
T1040	Network Sniffing
T1557	Adversary-in-the-Middle
T1110.002	Password Cracking
T1669	Wi-Fi Networks

These techniques represent realistic attacker behaviors observed during wireless intrusion campaigns.

Business Impact Assessment

Failure to address identified wireless security weaknesses may result in:

- Unauthorized wireless access
- Credential theft
- Loss of sensitive information
- Service disruption

- Compliance violations
- Increased incident response costs
- Reputational damage

Wireless security weaknesses can also serve as entry points for larger attacks against enterprise infrastructure.

Strategic Recommendations

The following actions are recommended:

Immediate Actions

- Strengthen wireless passwords
- Disable legacy wireless protocols
- Review wireless access policies
- Enable continuous monitoring

Short-Term Actions

- Deploy WPA3-Enterprise
- Implement NAC
- Deploy WIDS/WIPS
- Improve wireless logging

Long-Term Actions

- Implement Zero Trust architecture
- Deploy centralized wireless monitoring
- Enhance security awareness training
- Conduct periodic wireless security assessments

Executive Conclusion

The assessment demonstrated that wireless infrastructure remains a critical component of enterprise security and should be managed with the same level of attention as traditional network infrastructure.

While no successful unauthorized access was achieved during the assessment, several weaknesses were identified that could increase organizational risk if left unaddressed.

Implementation of the recommended remediation plan, improved monitoring capabilities, stronger authentication controls, and enhanced network segmentation will significantly improve the security posture of KAPAS INTERNATIONAL and reduce exposure to wireless-based threats.

Executive Risk Summary

Category	Status
Wireless Authentication	Needs Improvement
Network Segmentation	High Priority
Rogue AP Protection	Needs Improvement
Monitoring Capability	Needs Improvement
WPA3 Readiness	Recommended
Overall Wireless Security Posture	Moderate Risk

21. 30-60-90 Day Remediation Plan

The remediation plan provides a structured approach for improving the wireless security posture of KAPAS INTERNATIONAL following the findings identified during the wireless security assessment.

The plan is divided into three phases:

- First 30 Days (Immediate Actions)
- First 60 Days (Security Enhancement Actions)
- First 90 Days (Strategic Security Improvements)

The objective is to reduce wireless security risks while improving visibility, monitoring, authentication, and network segmentation.

First 30 Days – Immediate Remediation Actions

Objective

Address critical wireless security weaknesses identified during the assessment.

Activities

Strengthen Wireless Authentication

- Review existing wireless passwords
- Replace weak passwords with strong passphrases
- Enforce password complexity requirements

Disable Legacy Wireless Protocols

- Disable WPA
- Disable WEP
- Enforce WPA2 or WPA3 where supported

Review Wireless Access Policies

- Identify unauthorized users
- Remove inactive wireless accounts
- Review administrative access permissions

Baseline Wireless Environment

- Document all SSIDs
- Document all authorized access points
- Document wireless device inventory

Conduct Wireless Security Awareness Training

- Educate users on:
 - Rogue access points
 - Evil Twin attacks
 - Credential theft risks
 - Safe wireless usage

Expected Outcome

Reduction of immediate wireless security risks and improved visibility of wireless assets.

First 60 Days – Security Enhancement Actions

Objective

Implement stronger enterprise wireless controls.

Activities

Implement Network Segmentation

Deploy dedicated VLANs for:

- Corporate Users
- Guest Users
- BYOD Devices
- IoT Devices
- CCTV Infrastructure
- Administrative Systems

Deploy Network Access Control (NAC)

Implement:

- Device authentication
- Endpoint validation
- Access control enforcement

Improve Wireless Monitoring

Deploy:

- Wireless Intrusion Detection System (WIDS)
- Wireless Intrusion Prevention System (WIPS)

Improve Log Collection

Centralize logs from:

- Access Points
- Wireless Controllers
- RADIUS Servers
- Firewalls

Strengthen Guest Wireless Security

Implement:

- Captive Portal
- Guest Isolation
- Internet-only access policies

Expected Outcome

Improved wireless access control and reduced risk of unauthorized access.

First 90 Days – Strategic Security Improvements

Objective

Establish a mature enterprise wireless security program.

Activities

Migrate to WPA3-Enterprise

Implement:

- WPA3-Enterprise
- RADIUS Authentication
- Active Directory Integration
- Multi-Factor Authentication (MFA)

Implement Security Operations Monitoring

Integrate wireless logs into:

- SIEM Platform
- SOC Monitoring Workflows

Develop Wireless Incident Response Procedures

Create procedures for:

- Rogue AP incidents
- Deauthentication attacks
- Credential theft attempts

- Unauthorized wireless access

Conduct Wireless Penetration Testing Program

Schedule:

- Quarterly wireless assessments
- Annual penetration tests
- Continuous security reviews

Implement Zero Trust Principles

Apply:

- Least Privilege Access
- Continuous Verification
- Device Trust Validation

Expected Outcome

A resilient enterprise wireless infrastructure capable of resisting modern wireless attacks.

Remediation Roadmap Summary

Timeline	Priority	Key Actions
0–30 Days	Critical	Password review, protocol hardening, asset inventory, awareness training
31–60 Days	High	Segmentation, NAC, WIDS/WIPS, centralized logging
61–90 Days	Strategic	WPA3-Enterprise, MFA, SIEM integration, incident response program

Success Metrics

The following metrics should be used to measure remediation success:

- Reduction in unauthorized wireless connections
- Reduction in authentication failures
- Elimination of legacy wireless protocols
- Increased wireless visibility
- Faster incident detection times
- Improved compliance readiness
- Improved wireless security maturity

22. Final Conclusion

This project successfully demonstrated the process of conducting a comprehensive wireless infrastructure security assessment against a real-world wireless environment represented by **BET9A@IKAY SHOP** and mapped to the enterprise scenario of **KAPAS INTERNATIONAL**.

The assessment included wireless reconnaissance, wireless traffic analysis, WPA2 handshake capture assessment, Evil Twin attack evaluation, offline password attack testing, packet inspection, risk analysis, and MITRE ATT&CK mapping.

Tools utilized during the assessment included:

- Airodump-ng
- Airededdon
- Wireshark
- Kali Linux Wireless Utilities

The assessment confirmed that wireless networks remain attractive targets for attackers due to the potential for reconnaissance, credential attacks, rogue access points, deauthentication attacks, and unauthorized access attempts.

Although the offline password attack did not successfully recover the wireless password, the successful WPA2 handshake capture demonstrated that weak passwords could expose the organization to credential compromise through offline password-cracking techniques.

The findings were mapped to relevant MITRE ATT&CK techniques, including:

- T1595 – Active Scanning
- T1040 – Network Sniffing
- T1557 – Adversary-in-the-Middle
- T1110.002 – Password Cracking
- T1669 – Wi-Fi Networks

To address the identified risks, a redesigned enterprise wireless architecture, SOC monitoring strategy, risk register, and 30-60-90 day remediation plan were developed.

Implementation of the recommended controls will significantly improve wireless security, reduce organizational risk, strengthen authentication mechanisms, enhance monitoring capabilities, and improve the overall security posture of KAPAS INTERNATIONAL.

23. References

[1] MITRE ATT&CK Framework

<https://attack.mitre.org>

[2] Aircrack-ng Documentation

<https://www.aircrack-ng.org>

[3] Airgeddon Documentation

<https://github.com/v1s1t0r1sh3r3/airgeddon>

[4] Wireshark Documentation

<https://www.wireshark.org/docs>

[5] Kali Linux Documentation

<https://www.kali.org/docs>

24. Appendices

Appendix A – Assessment Screenshots

This appendix provides a reference list of all screenshots and assessment evidence collected during the wireless security assessment.

- Screenshot 01 – Wireless Adapter Detection
- Screenshot 02 – Monitor Mode Activation
- Screenshot 03 – Wireless Reconnaissance Results
- Screenshot 04 – Target Network Analysis
- Screenshot 05 – Airedddon Main Menu
- Screenshot 06 – Interface Selection
- Screenshot 07 – Monitor Mode Configuration
- Screenshot 08 – Target Selection
- Screenshot 09 – Handshake Attack Menu
- Screenshot 10 – Assessment Attempt Result
- Screenshot 11 – Successful Handshake Capture
- Screenshot 12 – Evil Twin Attack Menu
- Screenshot 13 – Offline Password Attack Result
- Screenshot 14 – Live Packet Capture
- Screenshot 15 – Target Focus Analysis
- Screenshot 16 – Packet Detail Analysis
- Screenshot 17 – Protocol Hierarchy Statistics
- Screenshot 18 – Wireless Traffic Observation

APPENDIX B – TOOLS USED DURING ASSESSMENT

Tool	Purpose
Airodump-ng	Wireless reconnaissance and target identification
Airedddon	Wireless assessment, handshake capture, Evil Twin testing
Wireshark	Packet capture and wireless traffic analysis
Kali Linux Wireless Utilities	Wireless interface management and monitor mode configuration

APPENDIX C – MITRE ATT&CK REFERENCES

MITRE ID	Technique	Tactic
T1595	Active Scanning	Reconnaissance
T1040	Network Sniffing	Credential Access, Discovery
T1557	Adversary-in-the-Middle	Credential Access, Collection
T1110.002	Password Cracking	Credential Access
T1669	Wi-Fi Networks	Initial Access

APPENDIX D – ABBREVIATIONS AND ACRONYMS

Acronym	Meaning
RF	Radio Frequency
SSID	Service Set Identifier
BSSID	Basic Service Set Identifier
AP	Access Point
WPA	Wi-Fi Protected Access
WPA2	Wi-Fi Protected Access Version 2
WPA3	Wi-Fi Protected Access Version 3
IoT	Internet of Things
BYOD	Bring Your Own Device
SOC	Security Operations Center
SIEM	Security Information and Event Management
NAC	Network Access Control
VLAN	Virtual Local Area Network
WIDS	Wireless Intrusion Detection System
WIPS	Wireless Intrusion Prevention System
MITRE	MITRE ATT&CK Framework
MFA	Multi-Factor Authentication

APPENDIX E – DISCLAIMER

Disclaimer

All activities, scans, exploitations, and simulations demonstrated in this repository were conducted in a controlled lab environment for educational and ethical purposes only. The target systems used were intentionally vulnerable systems owned or authorized for testing. Unauthorized testing against real-world systems is illegal and unethical.